# Capa de red

- Asignación de direcciones IP
- DNAT / SNAT
- ICMP
- IPv6

# Asignación de direcciones IP

Cada host en una red debe tener una dirección IP compatible con la red a la que pertenece, y a su vez dos hosts no pueden tener la misma dirección. Existen dos esquemas para asignación de direcciones

- **Direccionamiento estático**: se configura la interface con una dirección IP y máscara de red fija.
- **Direccionamiento dinámico**:
  - RARP ( RFC 903 )
  - BOOTP ( RFCs 951 y 1532 )
  - DHCP (RFCs 2131, 2132, upd: 3396, 3442, 3942)

# Asignación de direcciones IP

**BOOTP**

Es usado por un dispositivo cuando éste se inicia para conocer cuál será su dirección IP. Envía un mensaje broadcast a la red con su número MAC preguntando por su número IP. Un servidor BOOTP le informa cuál es el número de IP y le puede informar también la máscara de red, gateway y otros datos.

BOOTP es reemplazado por DHCP.

# DHCP (*Dynamic Host Configuration Protocol*)

- Es un sucesor de BOOTP
- El cliente puede conocer o no la dirección IP del servidor DHCP
- La asignación de IP es dinámica, aunque permite reservar direcciones IP para direcciones MAC
- Permite obtener, además del número IP
  - Máscara de red
  - Gateways ( IP )
  - DNS servers (nombre e IP)
  - etc. ( ver RFC 2132 )

# DHCP

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|     op (1)    |   htype (1)   |   hlen (1)    |   hops (1)    |
+---------------+---------------+---------------+---------------+
|                            xid (4)                            |
+-------------------------------+-------------------------------+
|           secs (2)            |           flags (2)           |
+-------------------------------+-------------------------------+
|                          ciaddr  (4)                          |
+---------------------------------------------------------------+
|                          yiaddr  (4)                          |
+---------------------------------------------------------------+
|                          siaddr  (4)                          |
+---------------------------------------------------------------+
|                          giaddr  (4)                          |
+---------------------------------------------------------------+
|                                                               |
|                          chaddr  (16)                         |
|                                                               |
|                                                               |
+---------------------------------------------------------------+
|                                                               |
|                          sname   (64)                         |
+---------------------------------------------------------------+
|                                                               |
|                          file    (128)                        |
+---------------------------------------------------------------+
|                                                               |
|                          options (variable)                   |
+---------------------------------------------------------------+
```

Ver RFC 2131 item 4.4

# DHCP: solicitud

DHCP server

IP: 200.1.2.3

**DHCP discover**

src: 0.0.0.0, 68

dest: 255.255.255.255, 67

yiaddr: 0.0.0.0

transaction ID: 456

**DHCP offer**

src: 200.1.2.3, 67

dest: 200.1.2.5, 68

yiaddr: 200.1.2.5

transaction ID: 456

Lifetime: 3600 secs

**DHCP request**

src: 0.0.0.0, 68

dest: 255.255.255.255, 67

yiaddr: 200.1.2.5

transaction ID: 456

Lifetime: 3600 secs

**DHCP ACK**

src: 200.1.2.3, 67

dest: 200.1.2.5, 68

yiaddr: 200.1.2.5

transaction ID: 456

Lifetime: 3600 secs

IP destino podría ser 255.255.255.255

# DHCP: ejemplo

Discover

```
▷ Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
◢ User Datagram Protocol, Src Port: 68, Dst Port: 67
    Source Port: 68
    Destination Port: 67
    Length: 308
    Checksum: 0x61a0 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
◢ Bootstrap Protocol (Discover)
    Message type: Boot Request (1)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0xfe089c15
    Seconds elapsed: 0
  ▷ Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 0.0.0.0
    Next server IP address: 0.0.0.0
    Relay agent IP address: 0.0.0.0
    Client MAC address: D-Link_12:47:cb (00:50:ba:12:47:cb)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  ▷ Option: (53) DHCP Message Type (Discover)
  ▷ Option: (116) DHCP Auto-Configuration
  ▷ Option: (61) Client identifier
  ▷ Option: (50) Requested IP Address
  ▷ Option: (12) Host Name
  ▷ Option: (60) Vendor class identifier
  ▷ Option: (55) Parameter Request List
  ▷ Option: (255) End
    Padding: 00000000
```

# DHCP: ejemplo

Offer

```
▷ Internet Protocol Version 4, Src: 10.20.20.4, Dst: 255.255.255.255
◢ User Datagram Protocol, Src Port: 67, Dst Port: 68
    Source Port: 67
    Destination Port: 68
    Length: 308
    Checksum: 0x8a44 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 2]
◢ Bootstrap Protocol (Offer)
    Message type: Boot Reply (2)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0xfe089c15
    Seconds elapsed: 0
  ▷ Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 10.20.20.20
    Next server IP address: 10.20.20.4
    Relay agent IP address: 0.0.0.0
    Client MAC address: D-Link_12:47:cb (00:50:ba:12:47:cb)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  ▷ Option: (53) DHCP Message Type (Offer)
  ▷ Option: (54) DHCP Server Identifier
  ▷ Option: (51) IP Address Lease Time
  ▷ Option: (1) Subnet Mask
  ▷ Option: (15) Domain Name
  ▷ Option: (6) Domain Name Server
  ▷ Option: (255) End
    Padding: 000000000000000000000000000000000000
```

# DHCP: ejemplo

Request

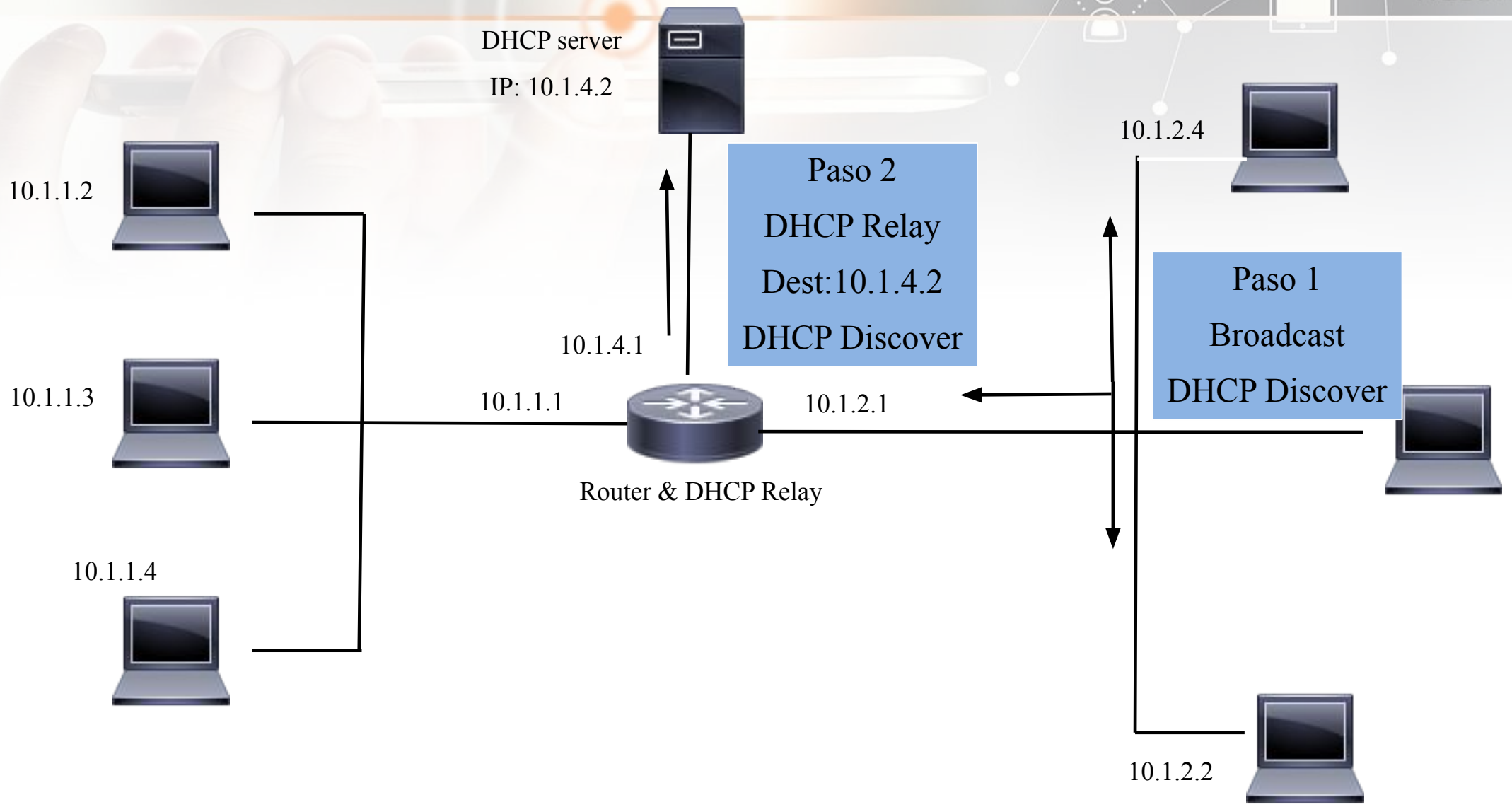
```
▷ Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
◢ User Datagram Protocol, Src Port: 68, Dst Port: 67
    Source Port: 68
    Destination Port: 67
    Length: 322
    Checksum: 0xf62b [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
◢ Bootstrap Protocol (Request)
    Message type: Boot Request (1)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0xfe089c15
    Seconds elapsed: 0
  ▷ Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 0.0.0.0
    Next server IP address: 0.0.0.0
    Relay agent IP address: 0.0.0.0
    Client MAC address: D-Link_12:47:cb (00:50:ba:12:47:cb)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  ▷ Option: (53) DHCP Message Type (Request)
  ▷ Option: (61) Client identifier
  ▷ Option: (50) Requested IP Address
  ▷ Option: (54) DHCP Server Identifier
  ▷ Option: (12) Host Name
  ▷ Option: (81) Client Fully Qualified Domain Name
  ▷ Option: (60) Vendor class identifier
  ▷ Option: (55) Parameter Request List
  ▷ Option: (255) End
```

# DHCP: ejemplo

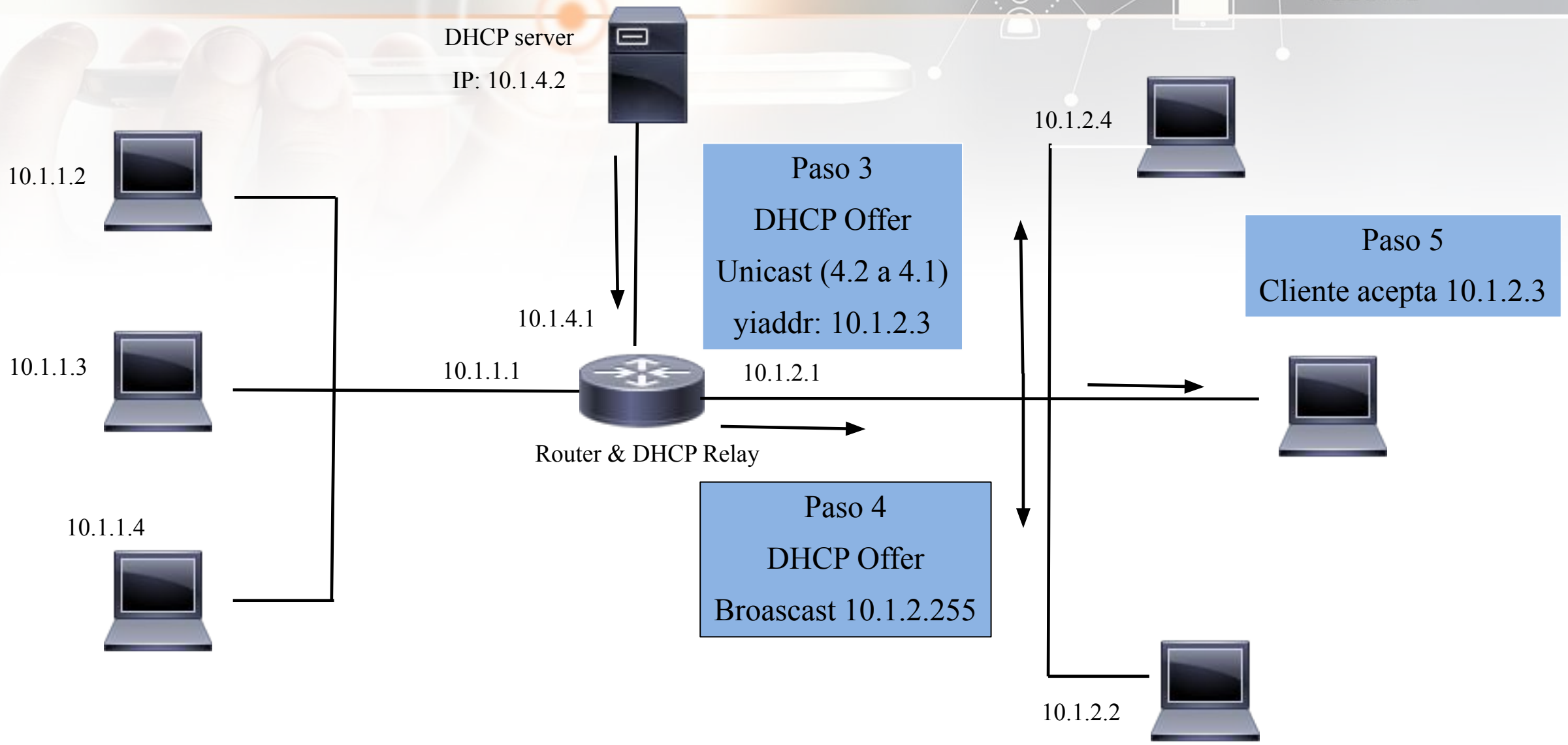
ACK

```
▷ Internet Protocol Version 4, Src: 10.20.20.4, Dst: 255.255.255.255
◢ User Datagram Protocol, Src Port: 67, Dst Port: 68
    Source Port: 67
    Destination Port: 68
    Length: 317
    Checksum: 0xb7a8 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 2]
◢ Bootstrap Protocol (ACK)
    Message type: Boot Reply (2)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0xfe089c15
    Seconds elapsed: 0
  ▷ Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 10.20.20.20
    Next server IP address: 10.20.20.4
    Relay agent IP address: 0.0.0.0
    Client MAC address: D-Link_12:47:cb (00:50:ba:12:47:cb)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  ▷ Option: (53) DHCP Message Type (ACK)
  ▷ Option: (54) DHCP Server Identifier
  ▷ Option: (51) IP Address Lease Time
  ▷ Option: (81) Client Fully Qualified Domain Name
  ▷ Option: (1) Subnet Mask
  ▷ Option: (15) Domain Name
  ▷ Option: (6) Domain Name Server
  ▷ Option: (255) End
```

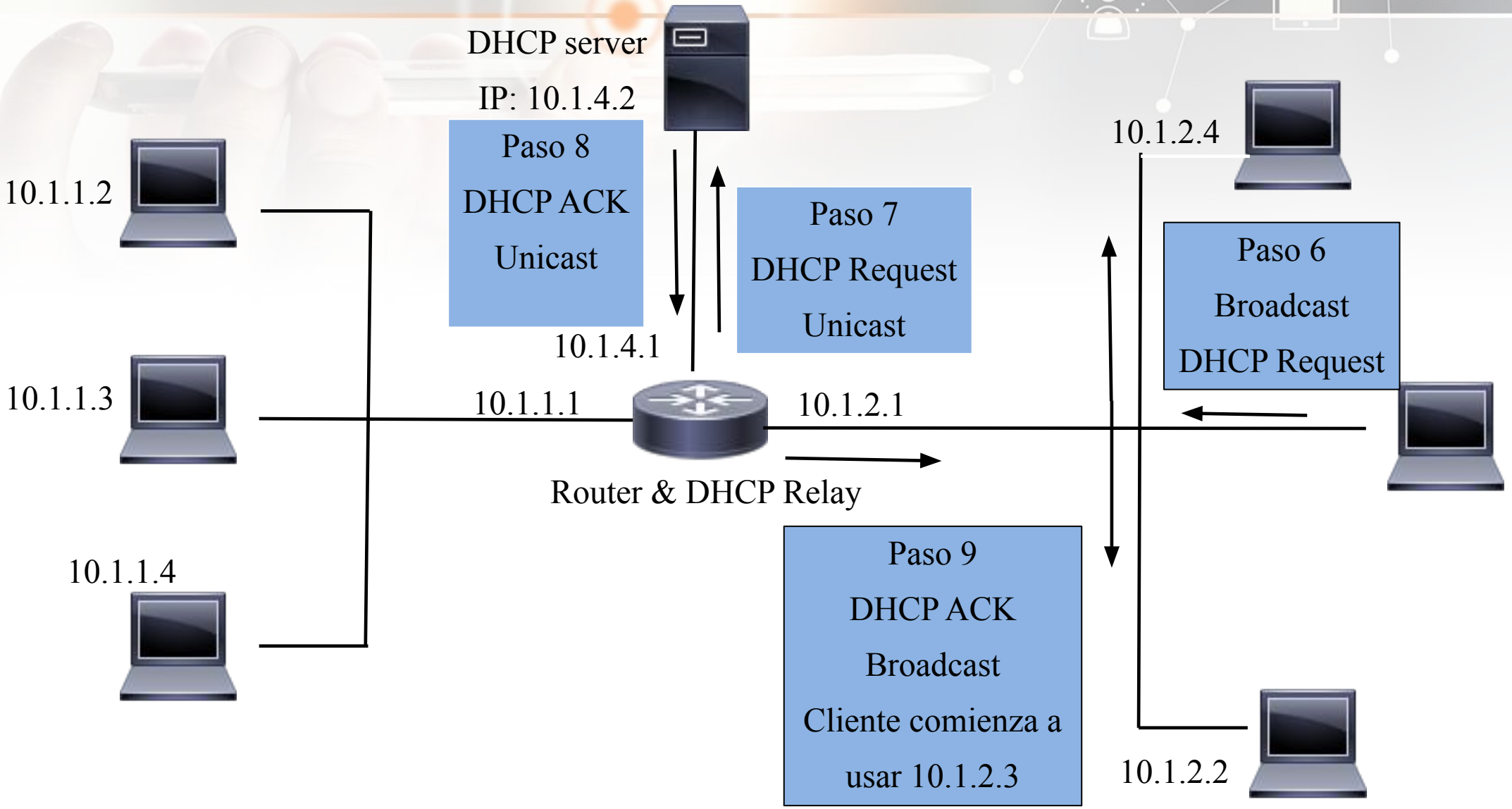
# DHCP: relay agent



# DHCP: relay agent



# DHCP: relay agent



# DHCP: renew request

```
▷ Internet Protocol Version 4, Src: 192.168.1.101, Dst: 192.168.1.1
▷ User Datagram Protocol, Src Port: 68, Dst Port: 67
◢ Bootstrap Protocol (Request)
    Message type: Boot Request (1)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0x0dd1b124
    Seconds elapsed: 0
  ▷ Bootp flags: 0x0000 (Unicast)
    Client IP address: 192.168.1.101
    Your (client) IP address: 0.0.0.0
    Next server IP address: 0.0.0.0
    Relay agent IP address: 0.0.0.0
    Client MAC address: AsustekC_22:77:a0 (40:16:7e:22:77:a0)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  ▷ Option: (53) DHCP Message Type (Request)
  ◢ Option: (61) Client identifier
      Length: 7
      Hardware type: Ethernet (0x01)
      Client MAC address: AsustekC_22:77:a0 (40:16:7e:22:77:a0)
```

```
  ▷ Option: (12) Host Name
  ▷ Option: (81) Client Fully Qualified Domain Name
  ▷ Option: (60) Vendor class identifier
  ◢ Option: (55) Parameter Request List
      Length: 12
      Parameter Request List Item: (1) Subnet Mask
      Parameter Request List Item: (15) Domain Name
      Parameter Request List Item: (3) Router
      Parameter Request List Item: (6) Domain Name Server
      Parameter Request List Item: (44) NetBIOS over TCP/IP Name Server
      Parameter Request List Item: (46) NetBIOS over TCP/IP Node Type
      Parameter Request List Item: (47) NetBIOS over TCP/IP Scope
      Parameter Request List Item: (31) Perform Router Discover
      Parameter Request List Item: (33) Static Route
      Parameter Request List Item: (121) Classless Static Route
      Parameter Request List Item: (249) Private/Classless Static Route (Microsoft)
      Parameter Request List Item: (43) Vendor-Specific Information
  ▷ Option: (255) End
```

# DHCP: renew ACK

```
▷ Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.101
▷ User Datagram Protocol, Src Port: 67, Dst Port: 68
◢ Bootstrap Protocol (ACK)
    Message type: Boot Reply (2)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0x0dd1b124
    Seconds elapsed: 0
  ▷ Bootp flags: 0x0000 (Unicast)
    Client IP address: 192.168.1.101
    Your (client) IP address: 192.168.1.101
    Next server IP address: 0.0.0.0
    Relay agent IP address: 0.0.0.0
    Client MAC address: AsustekC_22:77:a0 (40:16:7e:22:77:a0)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
```

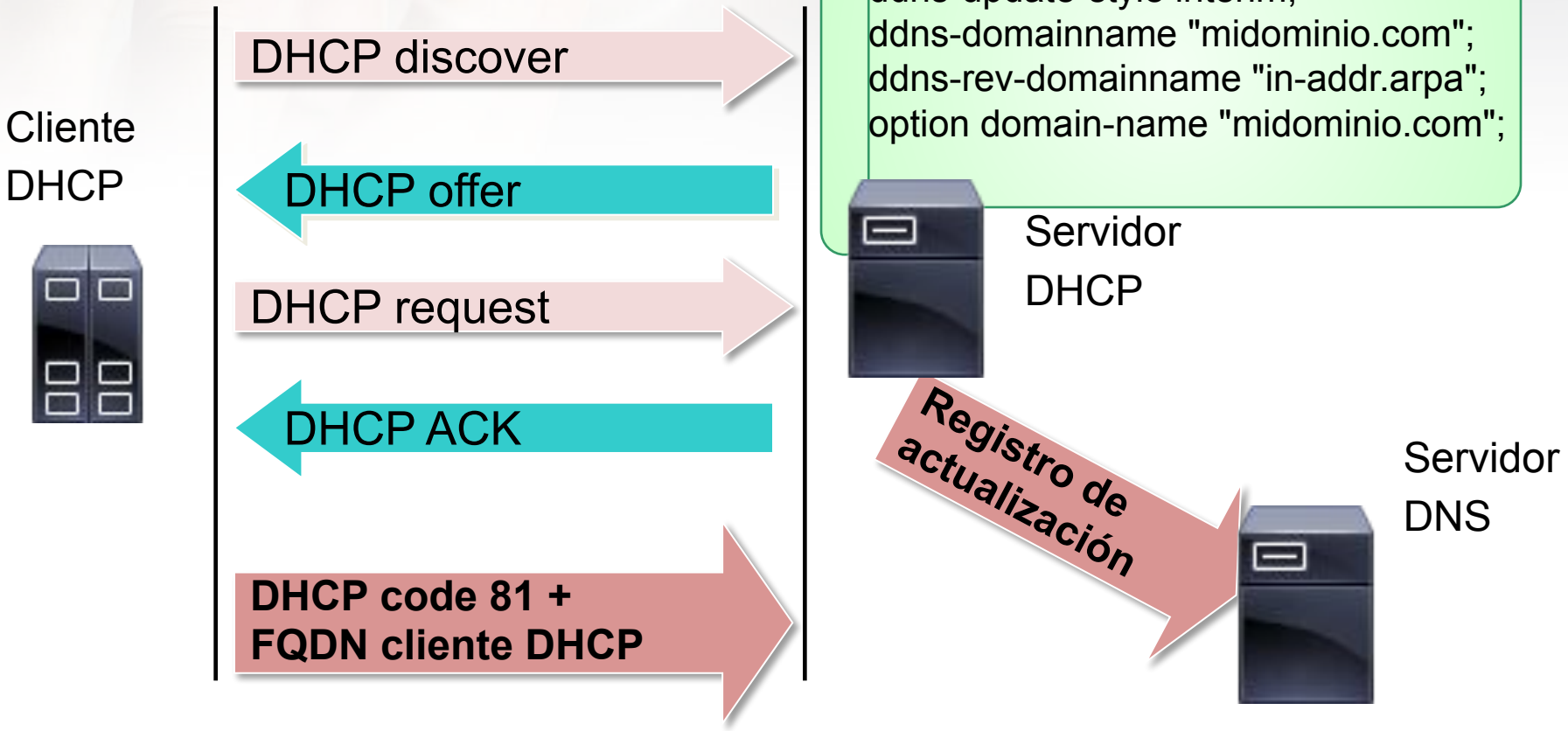
```
◢ Option: (53) DHCP Message Type (ACK)
      Length: 1
      DHCP: ACK (5)
◢ Option: (54) DHCP Server Identifier
      Length: 4
      DHCP Server Identifier: 192.168.1.1
◢ Option: (51) IP Address Lease Time
      Length: 4
      IP Address Lease Time: (4294967295s) infinity
◢ Option: (1) Subnet Mask
      Length: 4
      Subnet Mask: 255.255.255.0
◢ Option: (3) Router
      Length: 4
      Router: 192.168.1.1
◢ Option: (6) Domain Name Server
      Length: 8
      Domain Name Server: 208.67.222.222
      Domain Name Server: 8.8.8.8
◢ Option: (255) End
      Option End: 255
  Padding: 000000000000000000000000000000000000000000000000
```

# DNS dinámico

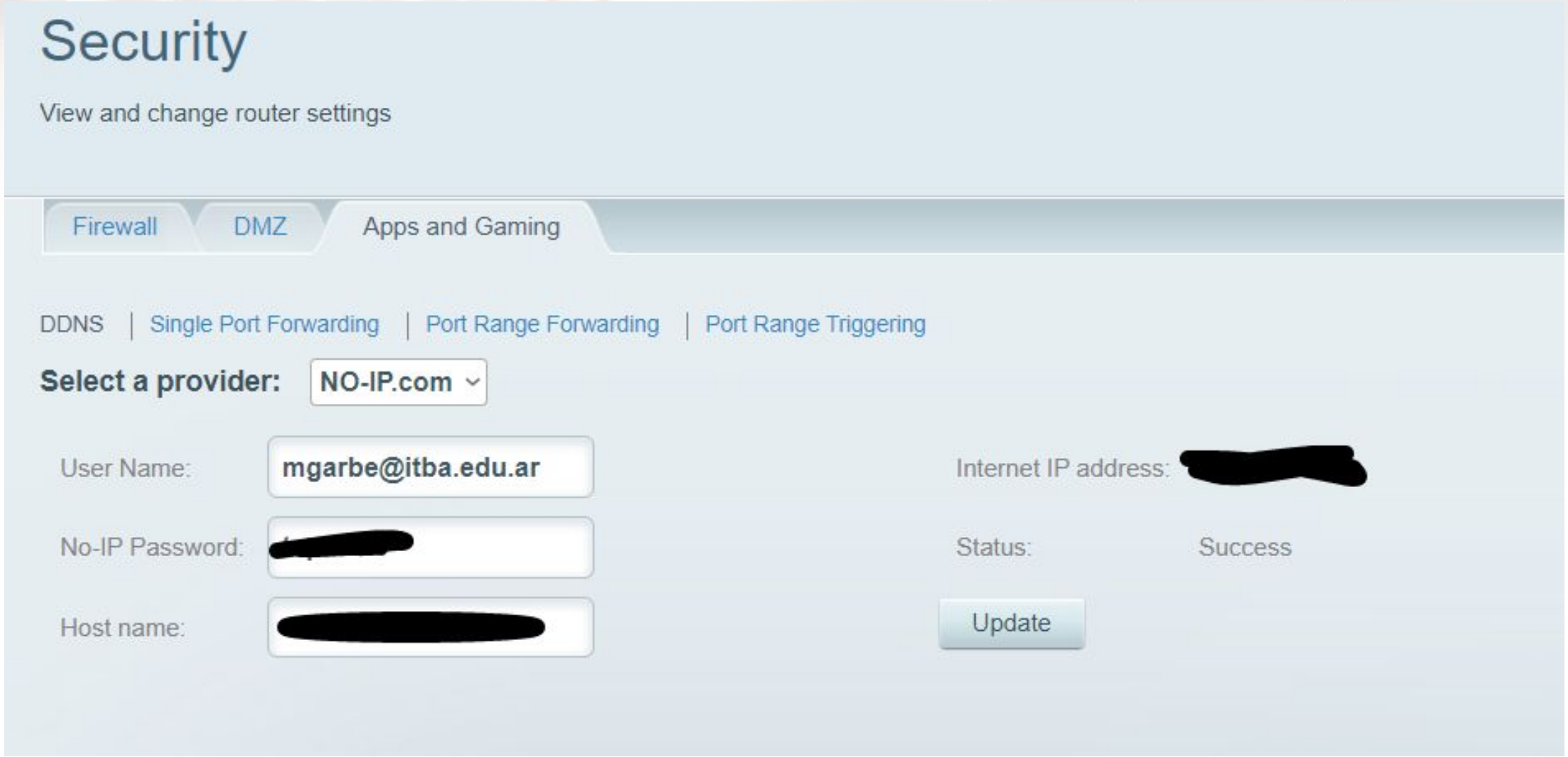
## Integración con DHCP

dhcp.conf

```
ddns-updates on;
ddns-update-style interim;
ddns-domainname "midominio.com";
ddns-rev-domainname "in-addr.arpa";
option domain-name "midominio.com";
```



# DDNS: ejemplo



# DHCP

DHCP es atendido por un servidor DHCP, no por un router

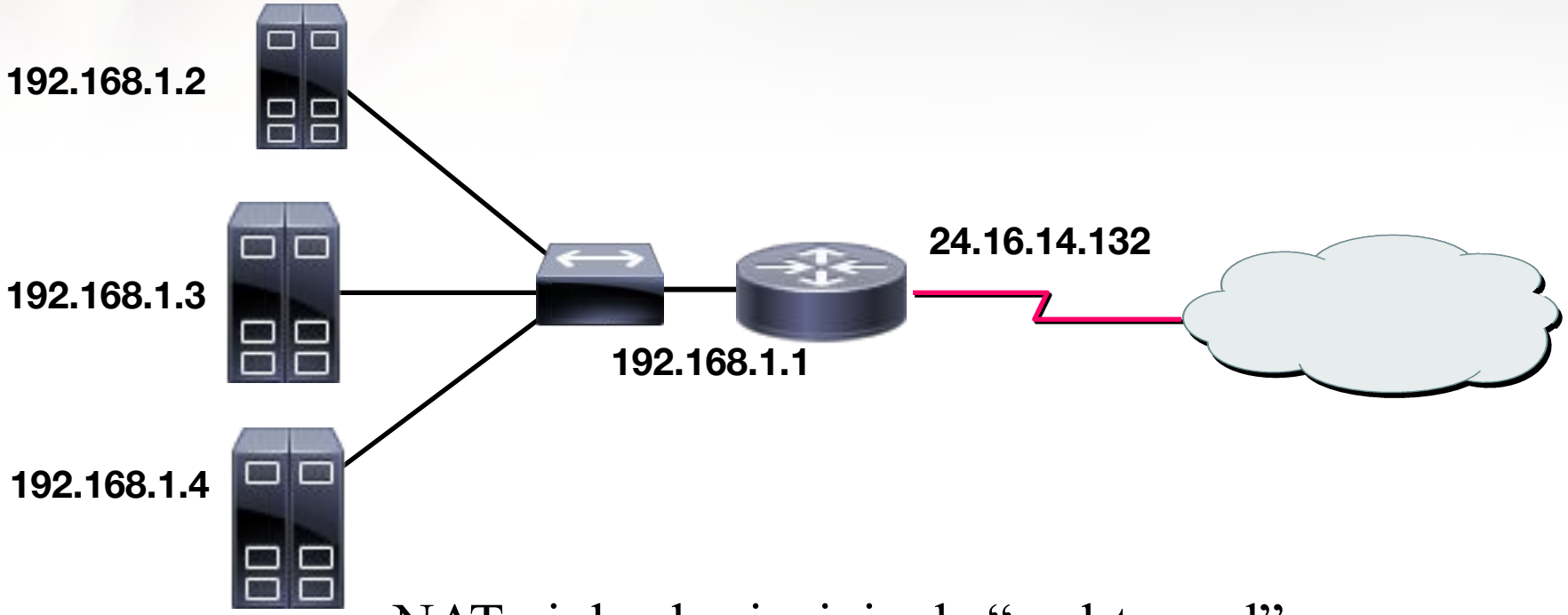
+

+ **DNS Caching-only Server**

+ **DHCP Server** + **Firewall**

# NAT (SNAT)

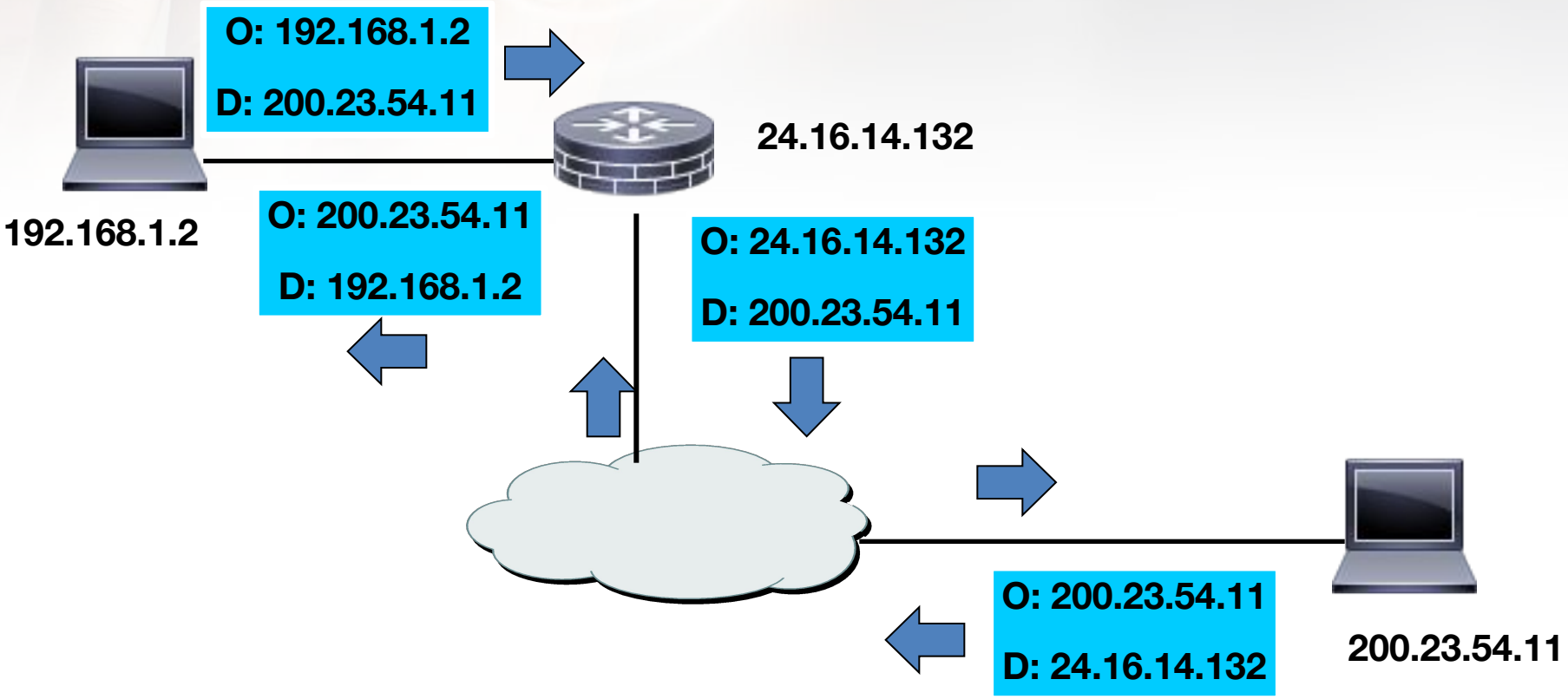
Una medida para aliviar la necesidad de IPs públicas fue **NAT** (1994, RFCs 1631 y 3022 )



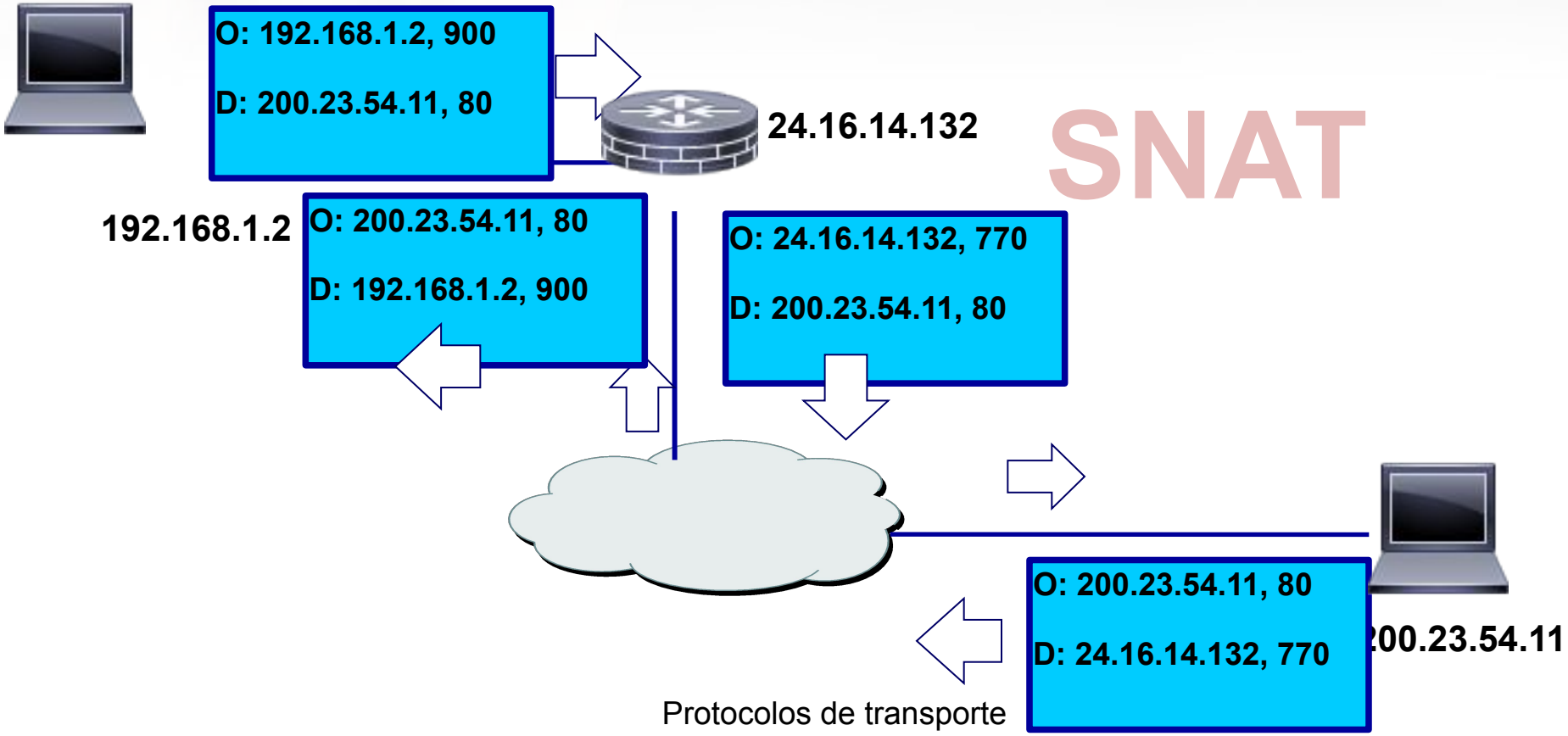

NAT viola el principio de “end-to-end”.
Debe ser tenido en cuenta por aplicaciones P2P

# SNAT

## Ejemplo



# NAPT (*aka* NAT)


O: 192.168.1.2, 900
D: 200.23.54.11, 80
24.16.14.132
SNAT
192.168.1.2
O: 200.23.54.11, 80
D: 192.168.1.2, 900
O: 24.16.14.132, 770
D: 200.23.54.11, 80
O: 200.23.54.11, 80
D: 24.16.14.132, 770
200.23.54.11
Protocolos de transporte

# NAT / NAPT

Tabla que mantiene el firewall para NAPT

| IP origen | Pto origen | IP destino | Pto destino | IP salida | Pto salida |
|---|---|---|---|---|---|
| 192.168.1.2 | 900 | 200.32.54.11 | 80 | 24.16.14.132 | 770 |
| 192.168.1.3 | 900 | 200.32.54.11 | 80 | 24.16.14.132 | 771 |

+

+ **DNS Caching-only Server**

+ **Firewall**

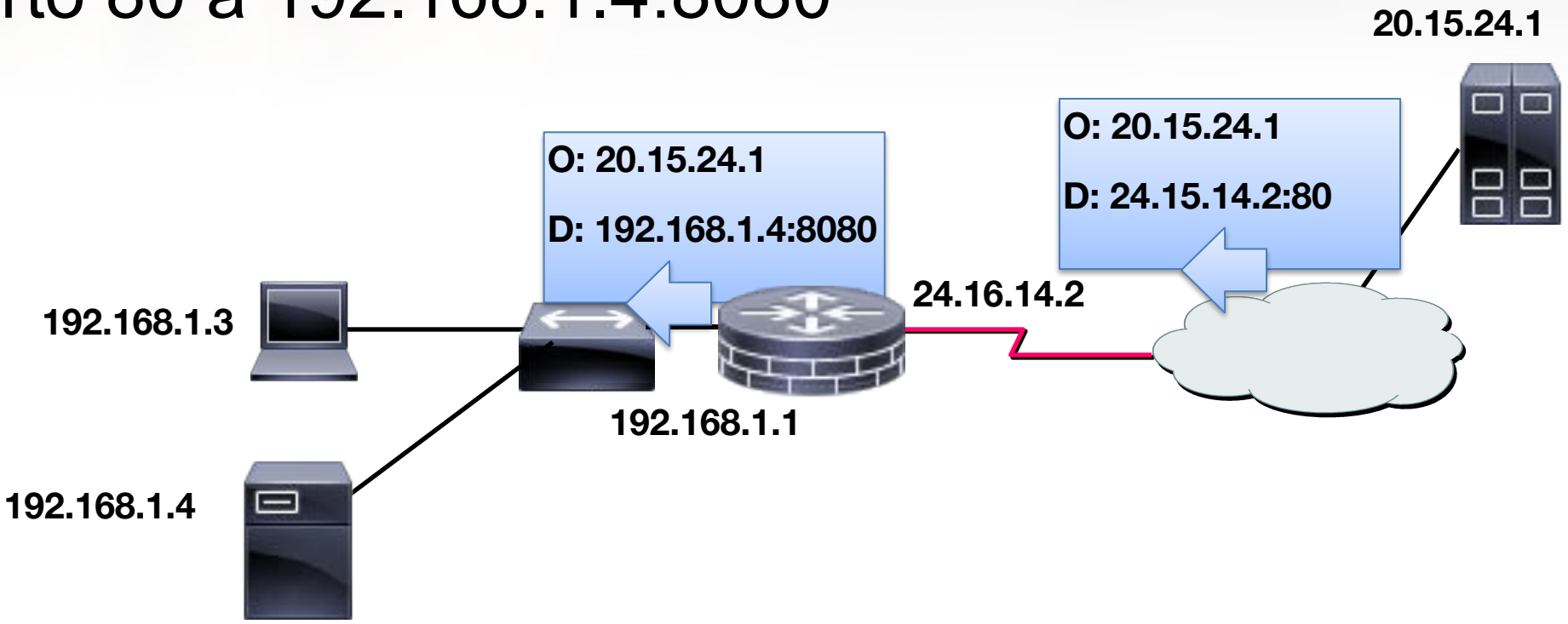
# SNAT

¿Qué campos se modifican al aplicar NAT a un paquete IP?

<table>
<tr><td>Vers</td><td>Hlen</td><td>ToS</td><td colspan="2">Longitud total</td></tr>
<tr><td colspan="3">Identificación</td><td>Flags</td><td>Desplazam. de fragmento</td></tr>
<tr><td>TTL</td><td colspan="2">Protocolo</td><td colspan="2">Header checksum</td></tr>
<tr><td colspan="5">Dirección origen</td></tr>
<tr><td colspan="5">Dirección destino</td></tr>
<tr><td colspan="4">Opciones</td><td>Relleno</td></tr>
<tr><td colspan="5">Datos</td></tr>
</table>

# DNAT (*port forwarding*)

- Quiero conectarme a un servidor en 192.168.1.4:8080
- configuro NAT para reenviar conexiones entrantes de puerto 80 a 192.168.1.4:8080



# DNAT: Ejemplo

## Security

View and change router settings

Firewall | DMZ | Apps and Gaming

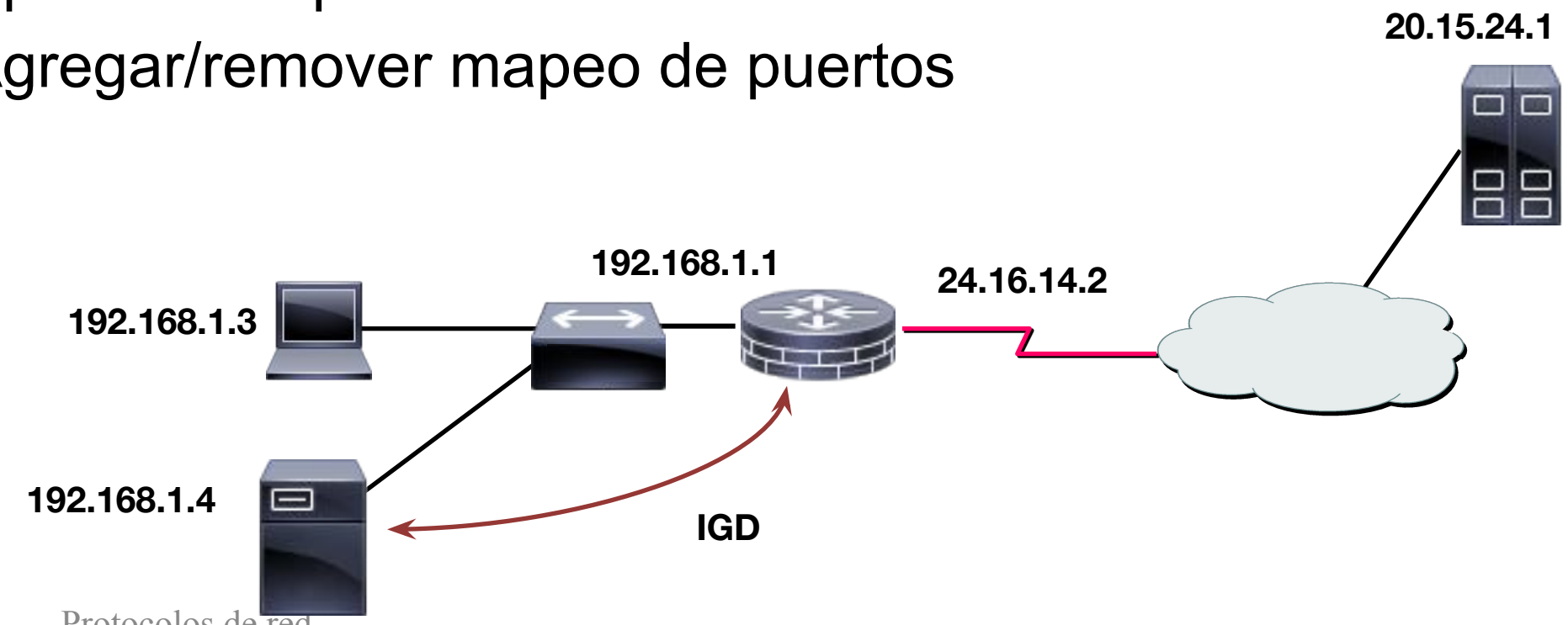
DDNS | Single Port Forwarding | Port Range Forwarding | Port Range Triggering

| Application name | External Port | Internal Port | Protocol | Device IP# | Enabled | |
|---|---|---|---|---|---|---|
| Apache | 9090 | 8080 | TCP | 192.168.1.142 | True | Edit/ Delete |
| Remoto Marce | 4489 | 3389 | Both | 192.168.1.127 | True | Edit/ Delete |
| Apache Crea | 9093 | 8080 | Both | 192.168.1.149 | True | Edit/ Delete |
| Front | 9100 | 8100 | Both | 192.168.1.149 | True | Edit/ Delete |
| Back | 9105 | 8105 | Both | 192.168.1.149 | True | Edit/ Delete |

Add a new Single Port Forwarding

# DNAT: port forwarding

- Solución 2: Universal Plug and Play (UPnP) Internet Gateway Device (IGD) Protocol. Permite al host "nateado"
  - Aprender IP pública
  - Agregar/remover mapeo de puertos



# UPnP

## UPnP

Note: Make sure the nat is enable if you want the UPnP configuration take effect

Current UPnP Status: Enabled Disable

### Current UPnP Settings List

| ID | App Description | External Port | Protocol | Internal Port | IP Address | Status |
|---|---|---|---|---|---|---|
| 1 | uTorrent (TCP) | 21103 | TCP | 21103 | 192.168.1.106 | Enabled |
| 2 | uTorrent (UDP) | 21103 | UDP | 21103 | 192.168.1.106 | Enabled |

Refresh

# NAT

Tanto DNAT como SNAT son funciones de un firewall, no de un router

+

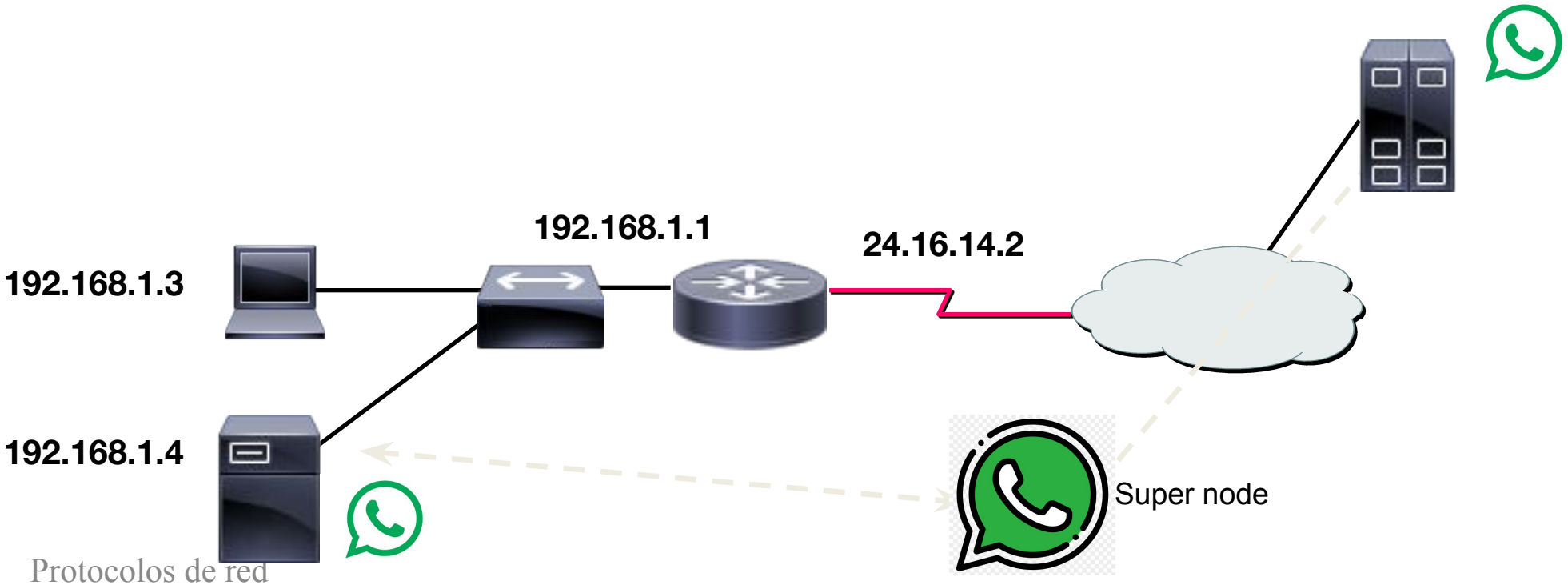
+ **DNS Caching-only Server**

+ **DHCP Server** +

+ **Webapp Server**

# NAT

- Solución 3: relaying (p.e. Skype, Whatsapp)
  - Host “nateado” conecta con “relay”
  - Host externo conecta con “relay”
  - El “relay” funciona como puente o proxy



# ICMP

**IP no maneja errores ni retransmisiones.**

**En caso de querer avisar de un error se debe usar mensajes ICMP**

| 0 | | | | | | | | | | 1 | | | | | | | | | | 2 | | | | | | | | | | 3 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0 | 1 |
| Type | | | | | | | | Code | | | | | | | | Checksum | | | | | | | | | | | | | | | |
| Data (Depende de Código y Dato) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

Los valores para los campos Tipo y Código se definen en RFC1700

# ICMP: Tipo y Código más comunes

| Tipo | Código | Descripción |
|---|---|---|
| 3 | 0 | Net unreachable |
| 3 | 1 | Host unreachable |
| 3 | 2 | Protocol unreachable |
| 0 | 0 | Echo reply |
| 8 | 0 | Echo request |
| 5 | 0 | Redirect datagrams for the Network |
| 5 | 1 | Redirect datagrams for the Host |
| 11 | 0 | TTL exceeded |
| 13/14 | 0 | Timestamp request/reply |

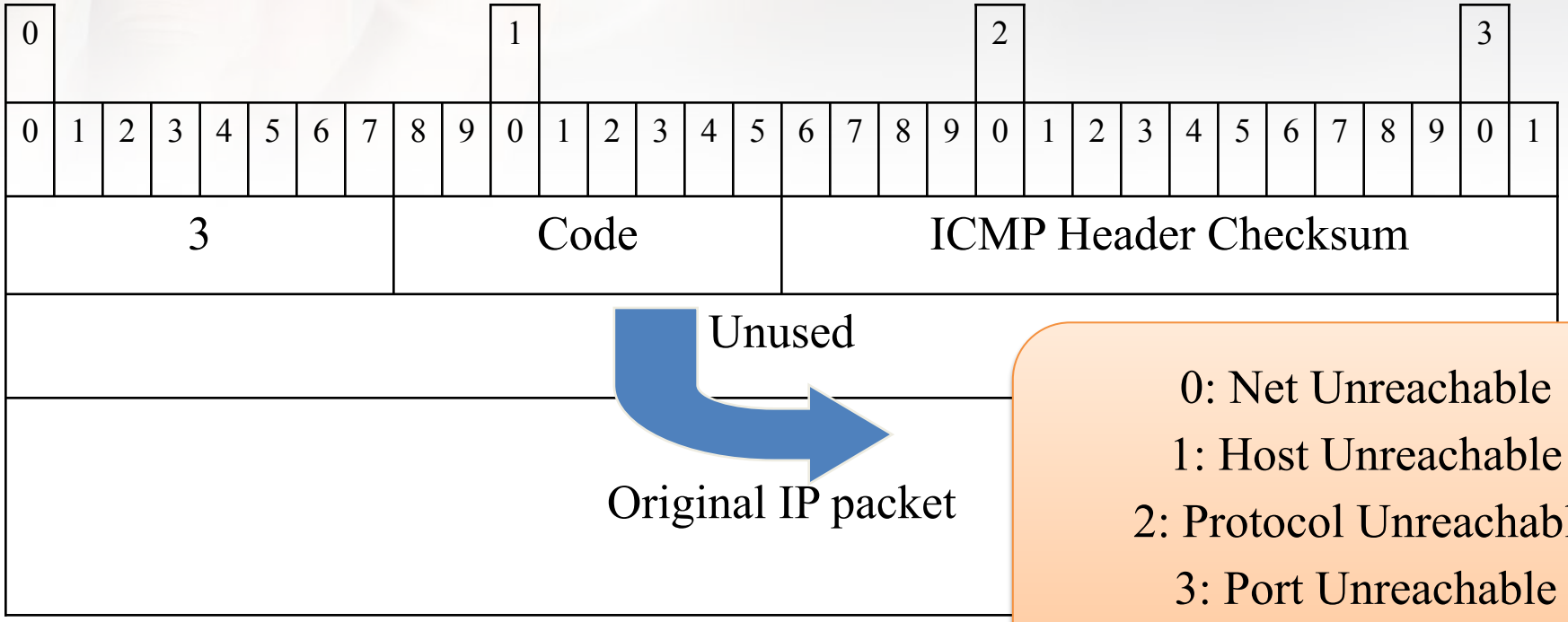
# ICMP: ejemplos

Timestamp request / reply

| 0 | | | | | | | | 1 | | | | | | | | 2 | | | | | | | | | | | | 3 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0 | 1 |
| 13/14 | | | | | | | | 0 | | | | | | | | ICMP Header Checksum | | | | | | | | | | | | | | | |
| Identifier | | | | | | | | | | | | | | | | Sequence number | | | | | | | | | | | | | | | |
| Originate timestamp | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Receive timestamp | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Transit timestamp | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

# ICMP: ejemplos

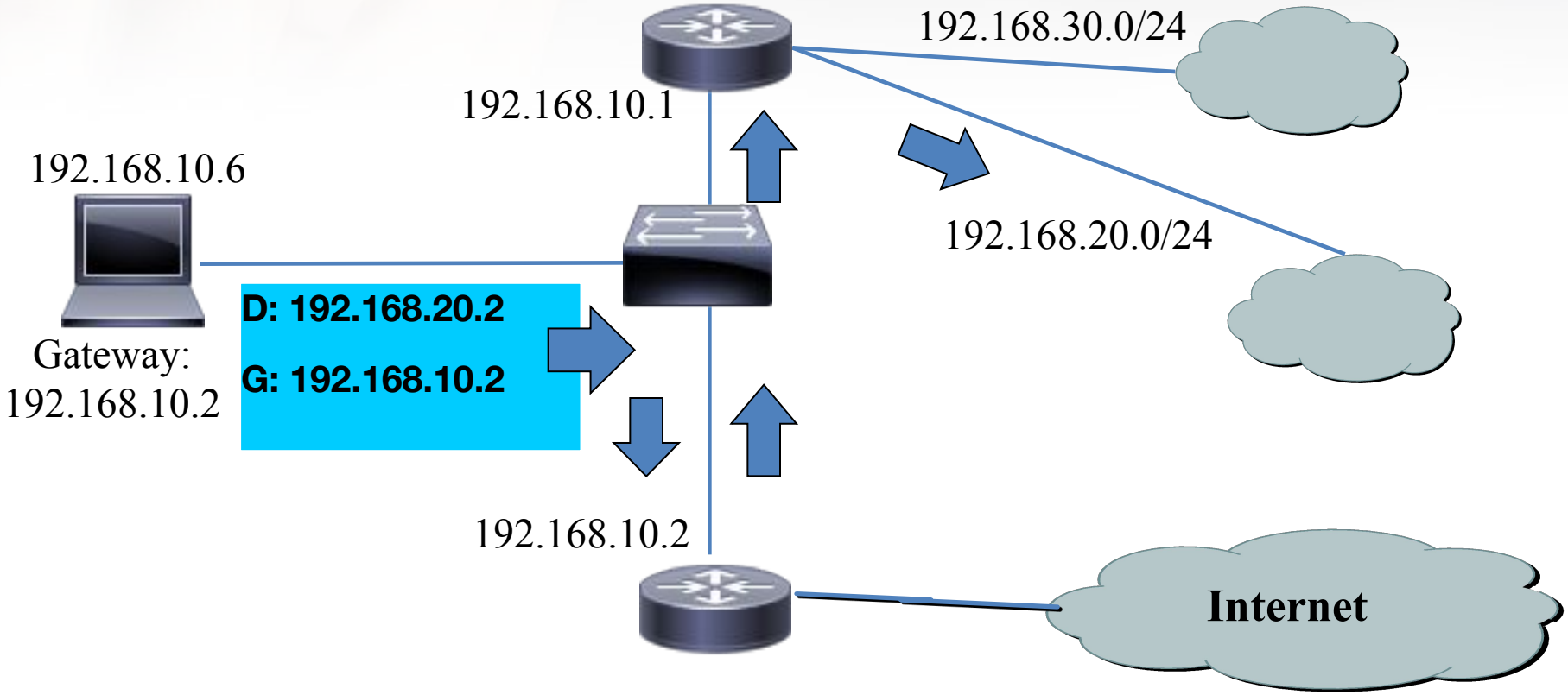
Destino inalcanzable




0: Net Unreachable
1: Host Unreachable
2: Protocol Unreachable
3: Port Unreachable
4: Fragmentation needed
…

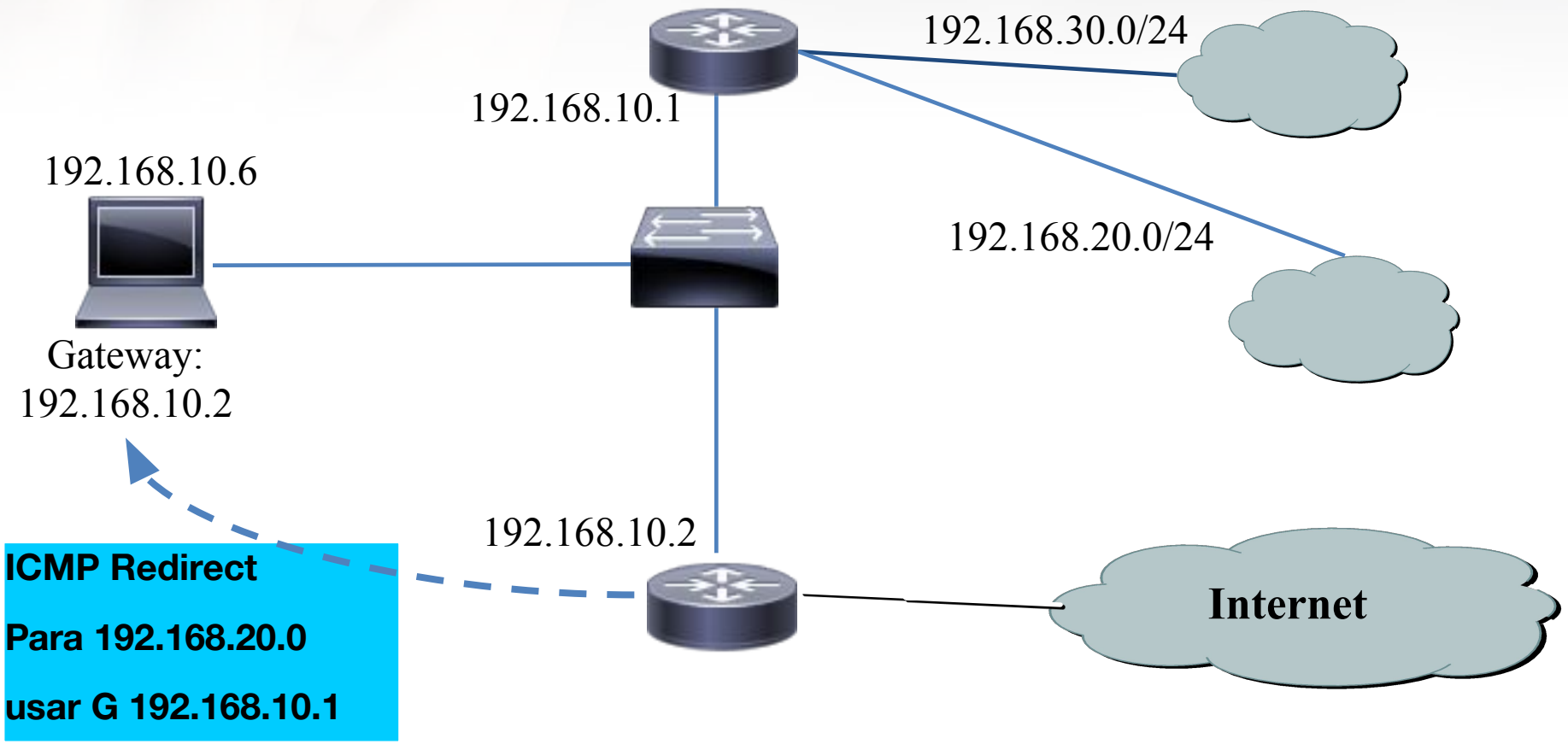
# ICMP: redirect

Las tablas de ruteo también se pueden actualizar en base a paquetes **ICMP Redirect**.



# ICMP: redirect

Las tablas de ruteo también se pueden actualizar en base a paquetes **ICMP Redirect**.



# Utilidades básicas de networking

**ping *hostDestino:*** Utiliza mensajes ICMP tipo 8 y 0

| 0 | | | | | | | | 1 | | | | | | | | 2 | | | | | | | | 3 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0 | 1 |
| 8 ó 0 | | | | | | | | 0 | | | | | | | | ICMP Header Checksum | | | | | | | | | | | | | | | |
| Identifier | | | | | | | | | | | | | | | | Sequence number | | | | | | | | | | | | | | | |
| Optional data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

Si *hostDestino* es alcanzable, mostrará un resumen de paquetes enviados, recibidos, perdidos y tiempo medio de transferencia.

# Utilidades básicas de networking

```
user@server:~$ ping 200.49.213.50

PING 200.49.213.50 (200.49.213.50) 56(84) bytes of data.
64 bytes from 200.49.213.50: icmp_seq=1 ttl=55 time=21.4 ms
64 bytes from 200.49.213.50: icmp_seq=2 ttl=55 time=37.2 ms
64 bytes from 200.49.213.50: icmp_seq=3 ttl=55 time=35.7 ms
64 bytes from 200.49.213.50: icmp_seq=4 ttl=55 time=25.2 ms
64 bytes from 200.49.213.50: icmp_seq=5 ttl=55 time=23.2 ms

--- 200.49.213.50 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4036ms
rtt min/avg/max/mdev = 21.424/28.601/37.273/6.594 ms
```

# Utilidades básicas de networking

```
user@server:~$ ping -c1 200.49.213.50 -r
PING 200.49.213.50 (200.49.213.50) 56(124) bytes of data.
64 bytes from 200.49.213.50: icmp_seq=1 ttl=56 time=2286 ms
RR:    192.168.2.10     209.13.133.250
    200.26.75.69     209.13.133.57
    200.51.241.170      200.51.240.82
    200.51.241.161      200.51.241.57
    172.30.248.46


--- 200.49.213.50 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 2286.637/2286.637/2286.637/0.000 ms
```

# Utilidades básicas de networking

En caso de tener problemas de conectividad, el comando **`ping`** nos puede ayudar a localizar dónde se encuentra el problema.

1. ping 127.0.0.1
2. ping *direccion_IP_local*
3. ping *host_en_mi_segmento*
4. ping *host_en_otro_segmento*
5. ping *host_fuera_de_la_red*

Tener en cuenta que muchos routers o hosts no responden solicitudes de eco.

# Utilidades básicas de networking

**traceroute**

Al igual que **`ping`**, trata de alcanzar un host destino, pero informa cada router por el que va pasando hasta llegar a destino.

Lo que hace en realidad es enviar mensajes **ICMP *echo request*** (igual que ping) pero primero con un tiempo de vida de 1 salto, luego de 2 saltos, 3 saltos, etc, con un máximo por defecto de 30 saltos.

Especialmente útil en dos casos:

- Detectar dónde “muere” un paquete
- Detectar dónde un paquete se “empantana”

# Traceroute: demoras reales

```
user@server:~$ traceroute campus.itba.edu.ar
Tracing route to campus.itba.edu.ar [192.230.225.58]
  1     <1 ms     <1 ms     <1 ms  192.168.1.1
  2      *         *         *     Request timed out.
  ...
  6     15 ms     15 ms     15 ms  129-161-89-200.fibertel.com.ar [200.89.161.129]
  7     13 ms     15 ms     16 ms  130-165-89-200.fibertel.com.ar [200.89.165.130]
  8     13 ms     14 ms     11 ms  222-165-89-200.fibertel.com.ar [200.89.165.222]
  9      *         *         *     Request timed out.
 10    139 ms    140 ms    138 ms  ae1-300G.ar5.MIA1.gblx.net [67.17.94.249]
 11      *         *         *     Request timed out.
 12      *         *         *     Request timed out.
 13    160 ms    163 ms    159 ms  BLACKBOARD.ear2.Washington1.Level3.net
[4.59.144.218]
 14    165 ms    159 ms    164 ms  eth4-1-bb1-dc2.blackboard.com [69.196.255.133]
 15    159 ms    169 ms    160 ms  eth4-3-bb1-va2.mhint [69.196.255.145]
 16    166 ms    174 ms    175 ms  eth3-3-gw1-va2.mhint [69.196.255.158]
 17    158 ms    159 ms    159 ms  itba.blackboard.com [192.230.225.58]
```

# IPv6

- Direcciones de 128 bits
- Estructura de direcciones jerárquica
- Número de MAC como parte de la dirección IP
- Incorpora seguridad y multicasting
- Encabezado de tamaño fijo, pensado para hacer más eficiente su procesamiento por los routers
- Facilita QoS
- No permite fragmentación

- El protocolo IPv6 se define en RFC 2460 ( Dic. 1998 )

# Estadísticas de uso

- https://www.internetsociety.org/deploy360/ipv6/statistics/
  - IPv6 – Google
  - IPv6 Measurement Maps

# Prefijos IPv6

| TIPO DE DIRECCIÓN | BITS DE PREFIJO | PREFIJO |
|---|---|---|
| Link-Local | 1111 1110 10 | FE80::/10 |
| Unspecified | 0000 ... 0 (128 bits) | ::/128 |
| Loopback | 0000 ... 01 (128 bits) | ::1/128 |
| Multicast | 1111 1111 | FF00::/8 |
| IPv4-Mapped | 000 ... 0111111111111 (96 bits) | ::FFFF/96 |
| ULA | 1111 110 | FC00::/7 |
| Global Unicast | 001 | 2000::/3 |
| Anycast | | |

# IPv6

- IPv4-compatible:
  - 0:0:0:0:0:FFFF:192.168.30.1
  - = ::FFFF:192.168.30.1
  - = ::FFFF:C0A8:1E01
- Como URL, se la encierra entre corchetes
  - http://[2001:1:4F3A::206:AE14]:8080/index.html
- No existe IP broadcast

# IPv6

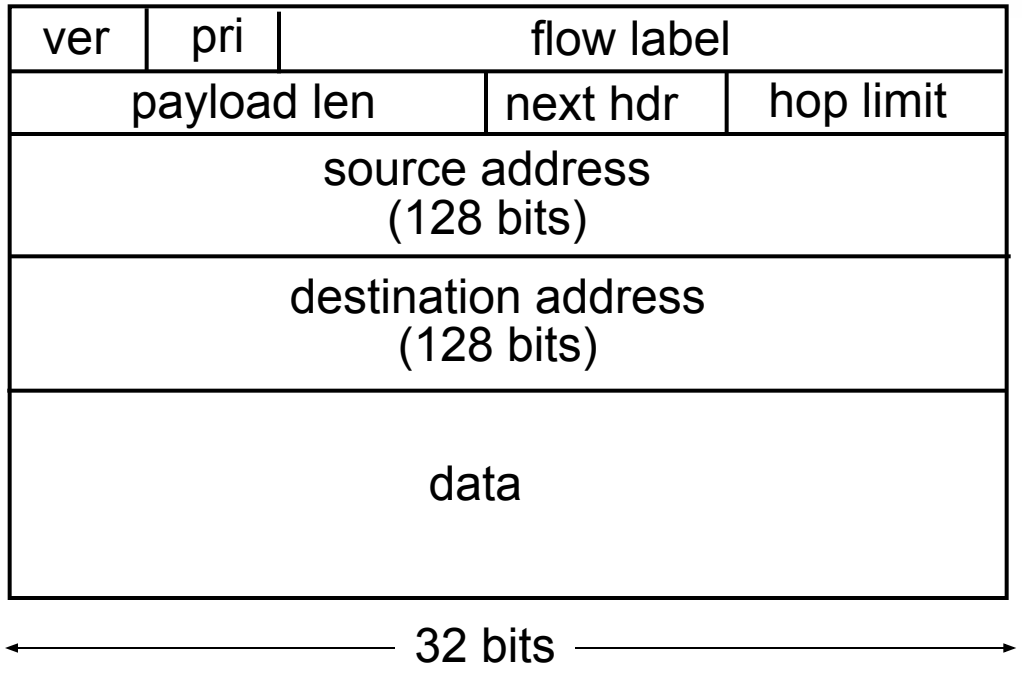
## Direcciones Global Unicast (RFC 3587)

| 3 | 13 bits | 32 bits | 16 bits | 64 bits |
|---|---|---|---|---|
| 001 | TLA ID | NLA ID | SLA ID | interface ID |
| | topología pública (*routing prefix*) | | topología del sitio (*subnet id*) | identificador de interfaz (*interface identifier*) |

*network prefix*

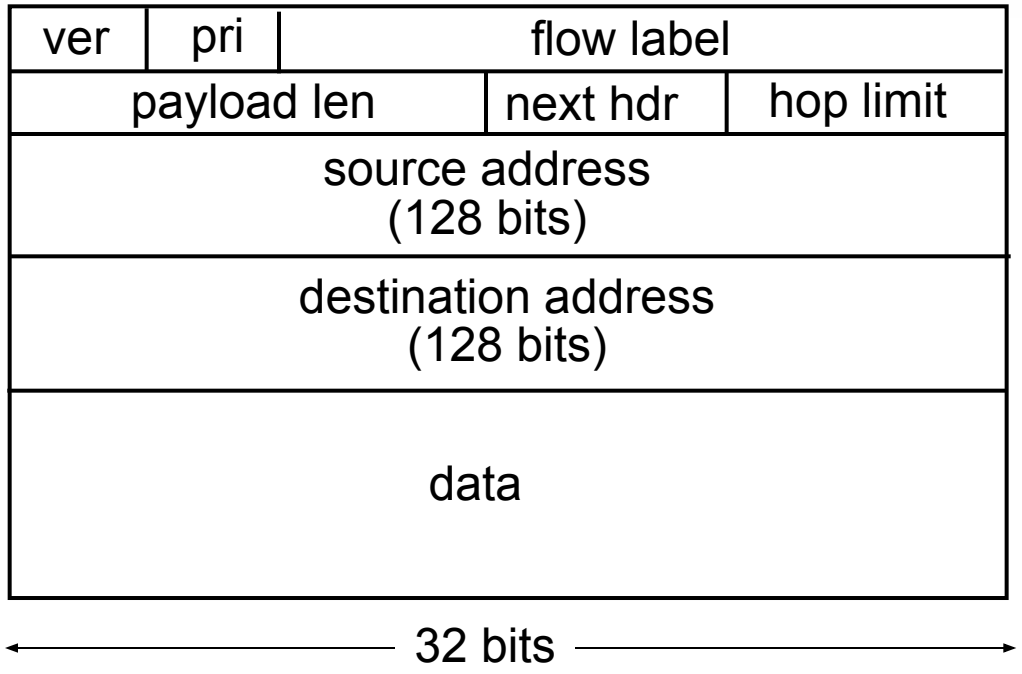
# IPv6: formato

- Priority: prioridad del paquete dentro del flujo
- Flow label: identifica paquetes dentro de un mismo “flujo”
- Checksum: no está más

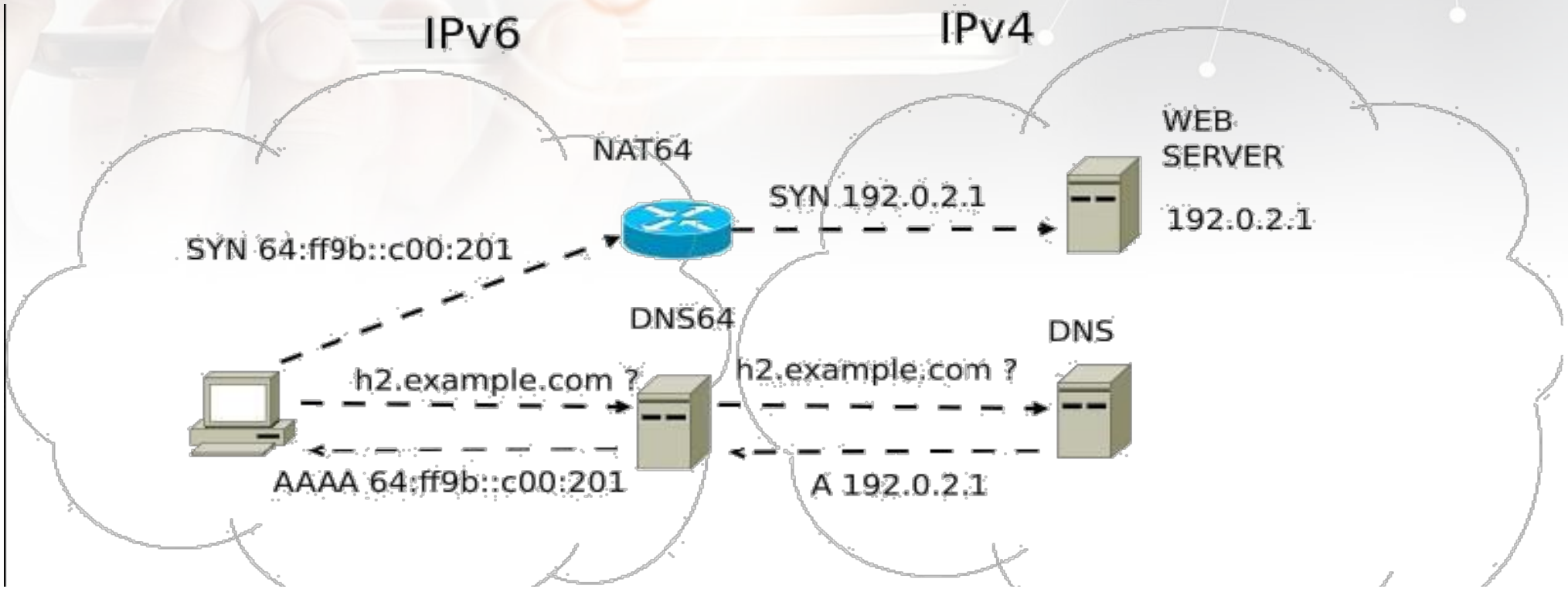


# IPv6: formato

- Payload len: longitud en octetos a continuación del header (hasta 64KB)
- Next header: similar a tipo de protocolo en IPv4
- Hop limit: similar a TTL en IPv4



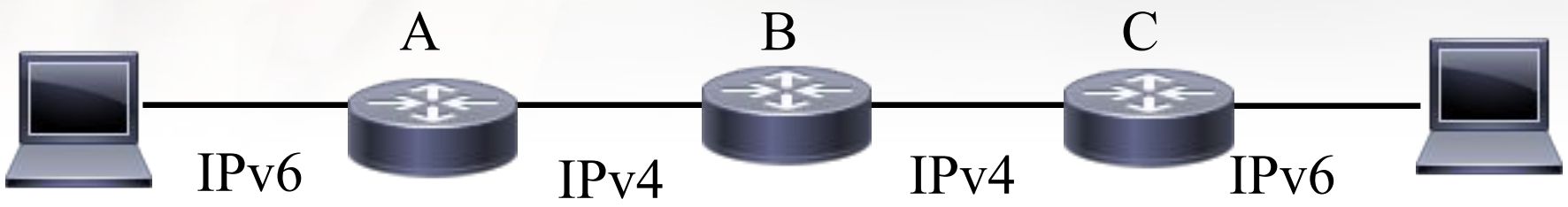
# NAT64: conectando IPv6 con IPv4




**NAT64** se encarga de la traducción de las direcciones entre hosts que sólo tienen conectividad IPv6 con hosts que sólo tienen IPv4. Tras la resolución del nombre por el DNS64, se pide la dirección web a través de NAT64.

# IPv6

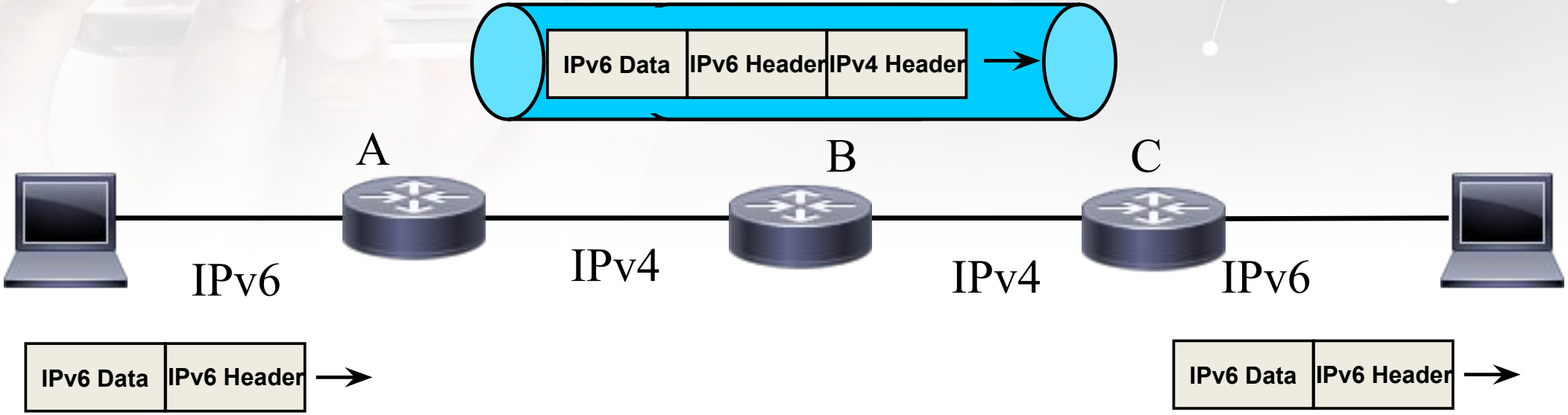
¿Qué sucede si una red intermedia sólo entiende IPv4?




El Router A debe "encapsular" el paquete Ipv6 dentro de un paquete IPv4

*Tunneling*: consiste en encapsular datos de un protocolo en los datos de otro protocolo de igual o mayor nivel.

# IPv6 over IPv4

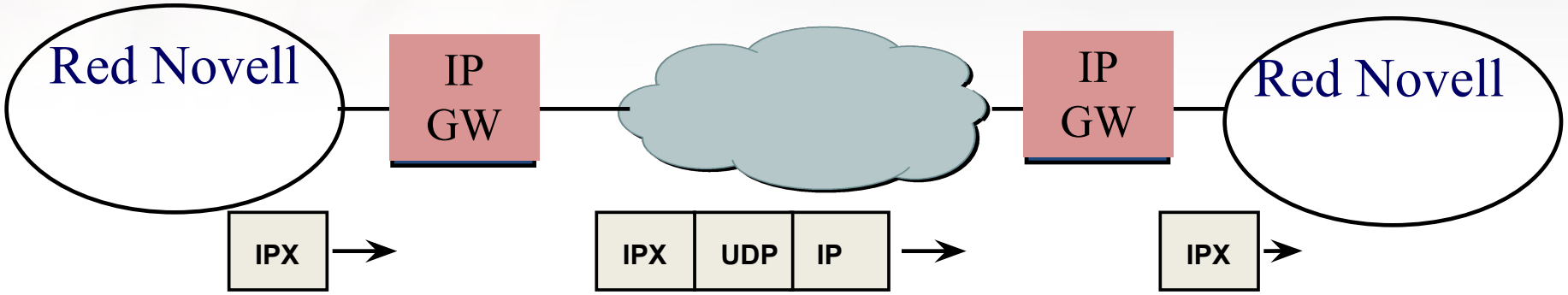


# Túneles

Supongamos que existen dos LAN Novell distantes que queremos conectar a través de Internet.

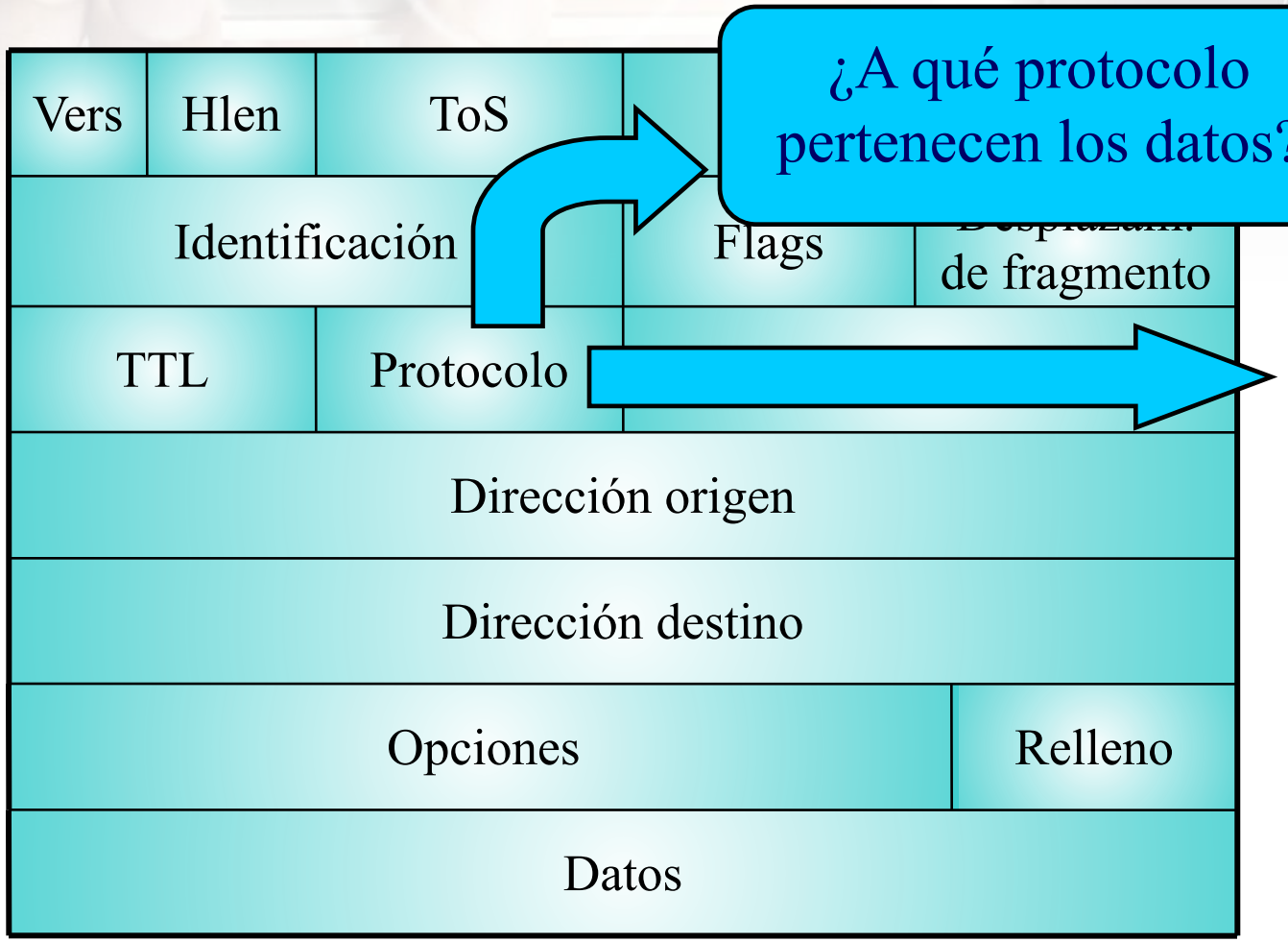
# Túneles

Solución: Encapsular los paquetes IPX en paquetes UDP



# Túneles IP-en-IP

Recordemos el paquete IP

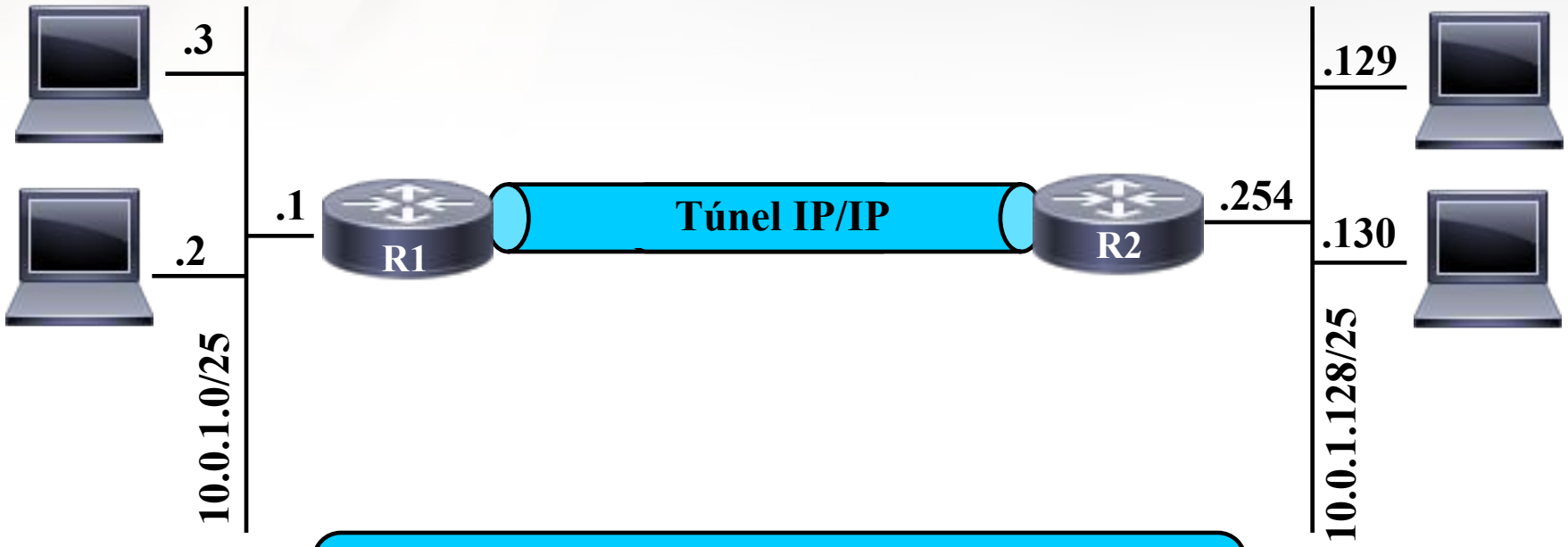



¿A qué protocolo pertenecen los datos?

| Valor | Prot. |
|---|---|
| 1 | ICMP |
| 2 | IGMP |
| 4 | IP |
| 6 | TCP |
| 17 | UDP |
| 41 | IPv6 |
| 89 | OSPF |

# Túneles IP-en-IP

Supongamos que tenemos dos redes IP y queremos enviar tráfico desde una hacia otra como si estuvieran conectadas directamente.




¿Podría resolverlo utilizando NAT?

# Túneles IP-en-IP

## Cómo configurar el túnel en Linux

En R1 (IP privada: 10.0.1.1, IP pública: 24.10.11.12)

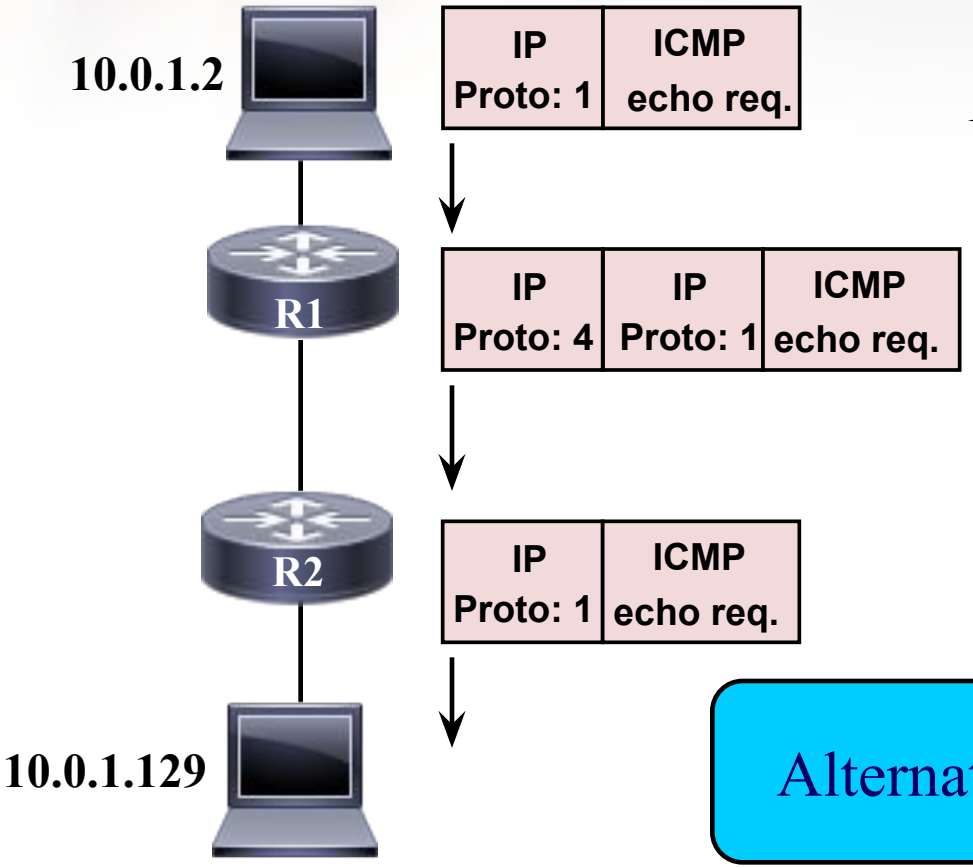
```
$ ifconfig tunl0 10.0.1.1 pointopoint 24.13.14.15
$ route add -net 10.0.1.128 netmask 255.255.255.128 dev tunl0
```

En R2 (IP privada: 10.0.1.254, IP pública: 24.13.14.15)

```
$ ifconfig tunl0 10.0.1.254 pointopoint 24.10.11.12
$ route add -net 10.0.1.0 netmask 255.255.255.128 dev tunl0
```

# Túneles IP-en-IP

¿Cuál sería el flujo de paquetes si el host 10.0.1.2 envía un ping al 10.0.1.129?




**¿Seguridad?**

Alternativa: protocolo PPP o IPSec

# Material de lectura

Capítulos 4.1 a 4.4 inclusive de la bibliografía

# IPSec

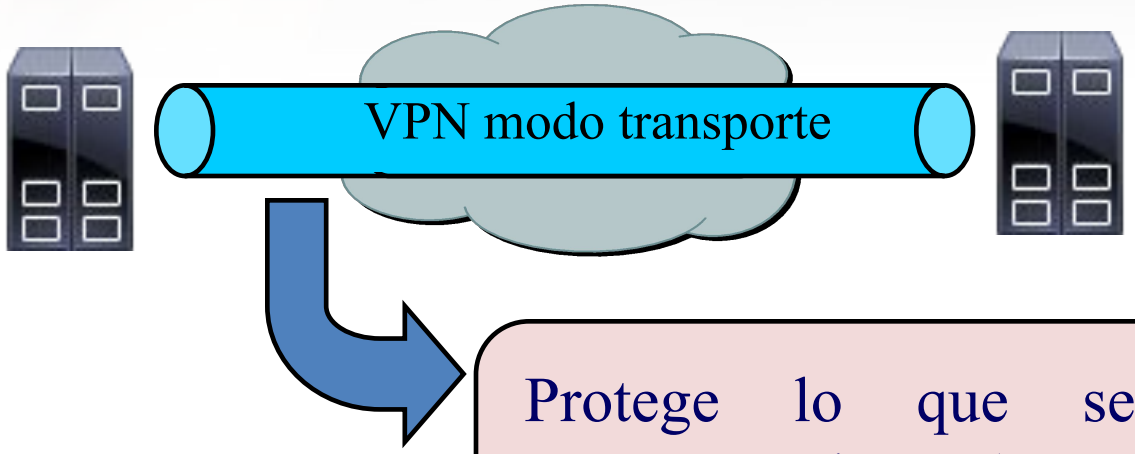
El objetivo de IPSec es proveer servicios de seguridad para los paquetes IP en el nivel de red.

- Estos servicios incluyen:
  - Control de acceso
  - Integridad de datos
  - Autenticación
  - Confidencialidad de datos

- IPSec trabaja en dos modos:
  - Túnel
  - Transporte

# IPSec: modo transporte

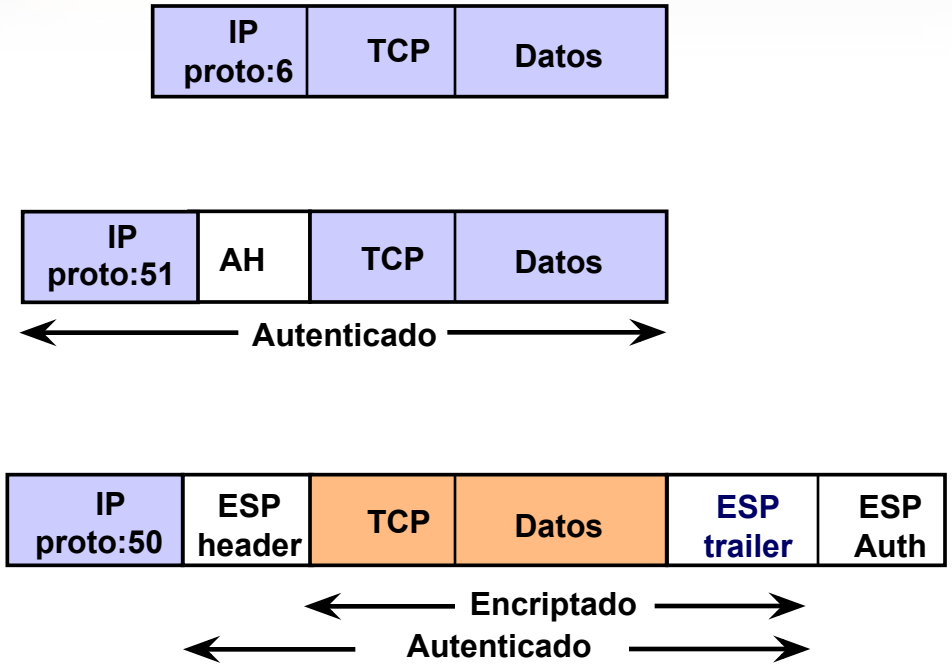
Es usado entre dos hosts en forma directa. No puede ser usado para conectar dos redes o un host a una red.




Protege lo que sea que esté transportando IP (TCP, UDP, ICMP, etc.).

# IPSec: modo transporte

En este modo se agrega un header AH (*Authentication Header*) o ESP (*Encapsulating Security Payload*)



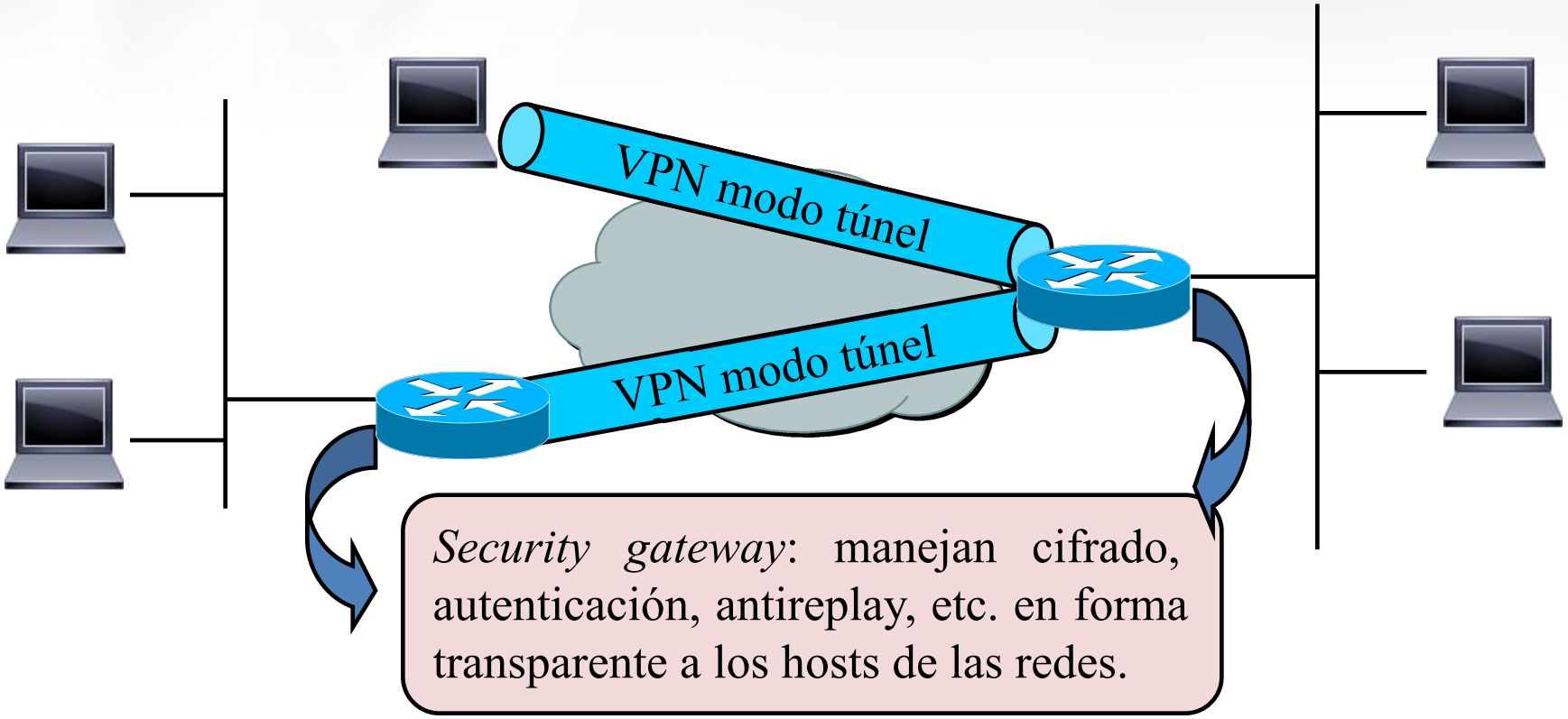

Paquete original

Autenticado

Autenticado y encriptado

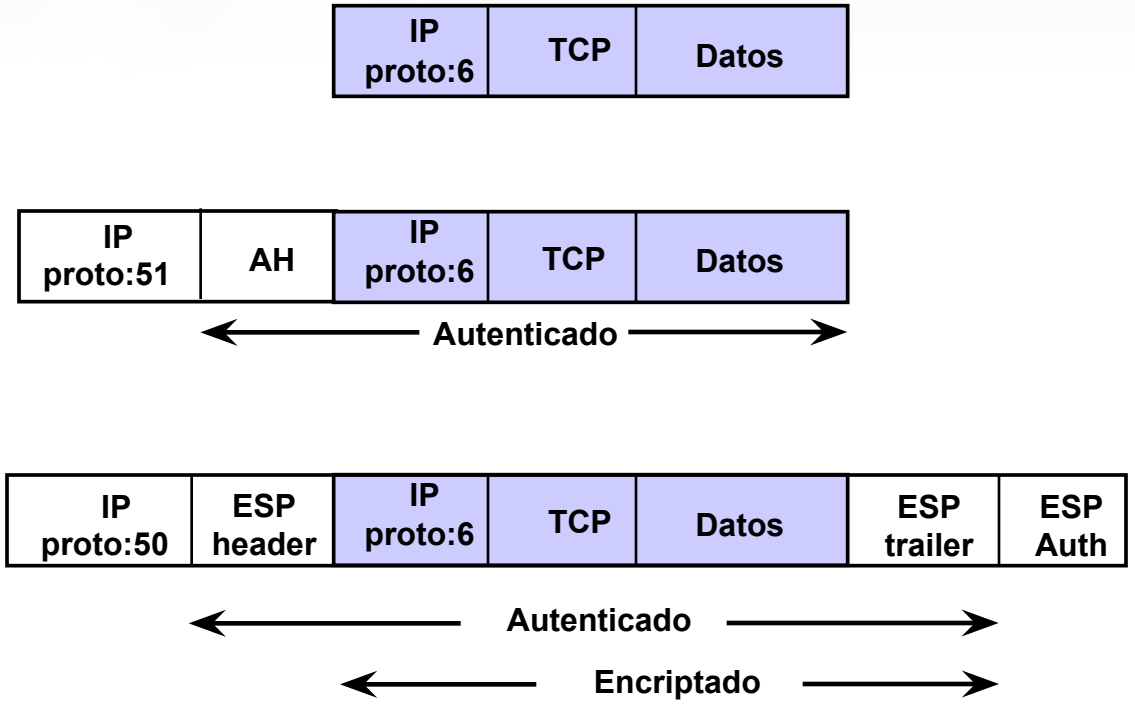
# IPSec: modo túnel

En modo túnel es posible formar una VPN con dos redes o un host y una red.



# IPSec: modo túnel

El paquete IP original es encapsulado en otro datagrama IP, insertando un header (AH o ESP)




Paquete original

Autenticado

Autenticado y encriptado

# IpTables e Ip6Tables

- Filtrado de paquetes y NAT
- Configurar, mantener e inspeccionar las tablas usadas por el kernel para filtrar paquetes
- Un paquete va pasando por distintas tablas, las cuales a su vez contienen una o más listas

# IPTables

Hay 5 tablas independientes que utiliza ipTables:

- **raw**: filtra los paquetes antes que otras tablas. Se utiliza principalmente para configurar exenciones de seguimiento de conexiones. Dos cadenas:
  - PREROUTING:
  - OUTPUT
- **filter**: es la usada por defecto. Cadenas:
  - INPUT: tráfico entrante
  - OUTPUT: tráfico saliente generado localmente
  - FORWARD: tráfico enrutado

```
~$ iptables -A INPUT -p tcp --dport 17500 -j REJECT
   --reject-with icmp-port-unreachable
```

# IPTables

Hay 5 tablas independientes que utiliza ipTables:

- **nat**: es consultada cuando un paquete crea una nueva conexión. Se usa para traducir direcciones de red y puertos. No se debe usar para reglas de filtrado. Cadenas:
  - PREROUTING: cambia paquetes antes de ser “ruteados”
  - POSTROUTING: cambia paquetes luego de ser “ruteados”
  - OUTPUT: aplica NAT a paquetes generados localmente
- Acciones:
  - DNAT
  - SNAT
  - MASQUERADE
  - REDIRECT

# Tabla nat

Utilizada para traducir el IP origen o destino de una secuencia de paquetes.

Posibles *targets*:

- DNAT
- SNAT
- MASQUERADE

```
~$ iptables -t nat -A PREROUTING -p tcp --dport 80
   -i eth1  -j DNAT --to 10.0.1.80:8080
```

# IpTables: Ejemplos

Listar reglas y ver tabla NAT actual

```
~$ iptables -t nat -L
~$ netstat-nat -n
```

Aplicar SNAT (eth1: interfaz WAN)

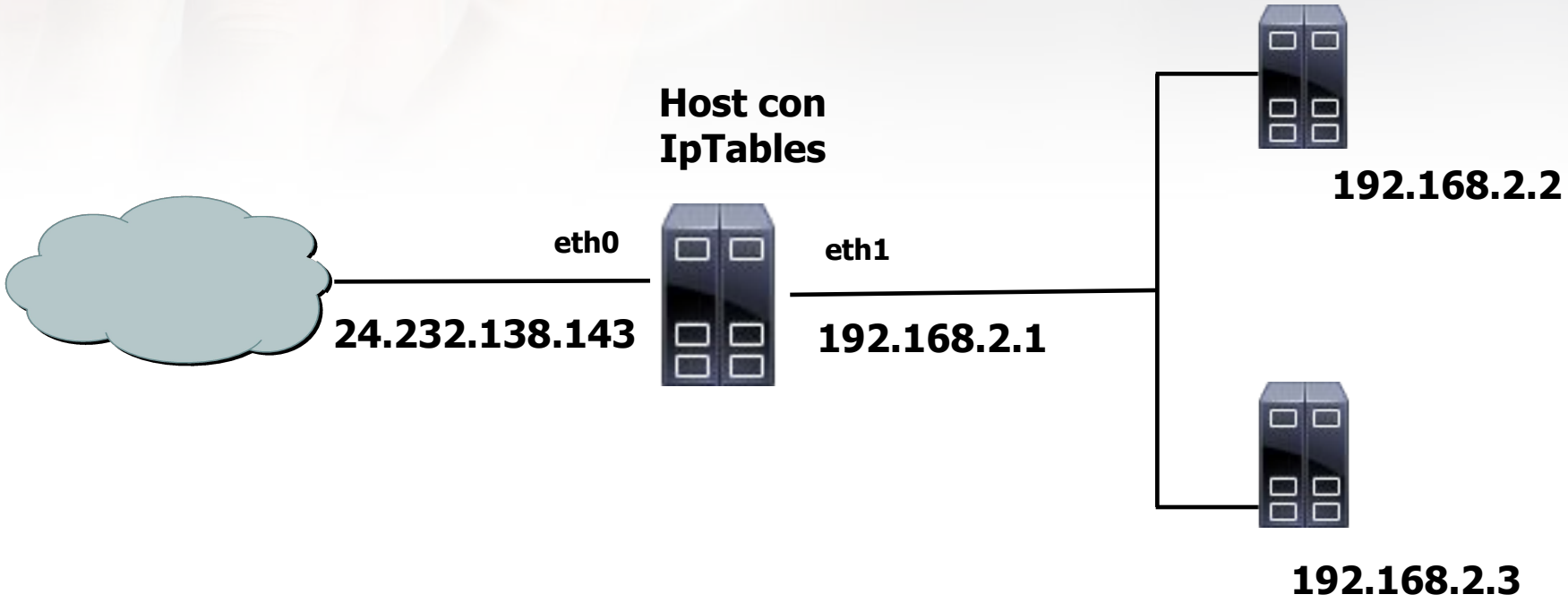
```
~$ iptables -t nat -A POSTROUTING -o eth1 -j MASQUERADE
```

```
~$ iptables -t nat -A POSTROUTING -s 192.168.1.0/24
   -o eth1 -j SNAT --to 20.28.101.14
```

```
~$ iptables -t nat -A POSTROUTING -p tcp
   -o eth1 -j SNAT --to 20.28.101.14:1-1023
```

# Ejemplo

Sea el siguiente escenario



# Ejemplo

```
# Reenviar todos los paquetes de eth1 (red interna)
# a eth0 (Internet).
iptables -A FORWARD -i eth1 -o eth0 -j ACCEPT

# Habilitar SNAT  para IP pública fija:
iptables -t nat -A POSTROUTING -o eth0 -j SNAT
         -to-source 24.232.138.143

# No aceptar IP Spoofing
iptables -A INPUT -i eth0 -s 24.232.138.143/32  -j DROP
iptables -A INPUT -i eth0 -s 192.168.0.0/24  -j DROP
iptables -A INPUT -i eth0 -s loopback/8 -j DROP
```

# Ejemplo

```
# Aceptamos lo que venga de localhost
iptables -A INPUT -i lo -j ACCEPT

# Implementamos Web Server en un host local
iptables -t nat -A PREROUTING -d 24.232.138.143
  -p tcp -dport 80 -j DNAT
  --to-destination 192.168.2.3
```